

The Long Lasting Effects of Warfare on Tax Progressivity: Evidence from World War II*

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Abstract

This paper studies how World War II reshaped progressive taxation. Treating the war as a natural experiment in a panel of 35 countries from 1929 to 2010, we estimate that entering the war raised a country's top marginal income tax rate by about 32 percentage points within five years, an effect that stayed large for three decades and statistically distinct from zero for about four before fading after 1980. Progressivity also rose on measures beyond the statutory rate: the average rate paid at the top, the tax on inheritances, and the compression of incomes by taxes and transfers. We interpret the persistence as a shift between political-economic equilibria: by compressing the wealth distribution, the war induced a new equilibrium that kept top taxation high for decades.

JEL codes: D31, D72, H24, N40

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1 Introduction

Why did top marginal income tax rates rise so steeply in some countries in the mid-twentieth century, and why did those increases last as long as they did? In the countries that mobilized for World War II, top rates climbed to historic highs in the 1940s and stayed elevated for several decades, narrowing toward the rates of countries that did not mobilize only much later (Scheve and Stasavage, 2016). Both the size of the rise and its timing point to the war itself. We argue that this wartime shock widened the divergence in how progressively these countries taxed and held it wide for decades.

The main empirical finding of this paper is that World War II raised top marginal tax rates sharply and durably in the countries that mobilized for it. Treating the war as a natural experiment, we estimate its effect on top rates in a panel of 35 countries from 1929 to 2010.¹ Entering the war changed more than tax law: it brought mass mobilization and, with it, lasting political and institutional change. We estimate the effect of the event as a whole. Five years after entry, a war-active country's top rate is about 32 percentage points higher than the no-war path;² ten years out, about 35; three decades out, about 33. The effect stays large for roughly three decades and remains statistically distinct from zero for about four, before fading, coinciding with the broad fall in top rates across countries after 1980. The size of the effect moves across the alternative specifications in Section 4, but its shape does not: a sharp rise at entry, and a top rate still above the no-war path three decades on.

Two forces connect mass-mobilization warfare to higher top rates. The first is fiscal: waging a mass-mobilization war required revenue on a scale pre-war tax systems could not supply (Harrison, ed, 1998), and the income tax was the instrument governments turned to (Obinger and Petersen, 2017). The second is political: as states conscripted ordinary citizens to fight, democratic publics demanded that owners of wealth bear a comparable burden, a conscription of wealth to match the conscription of labor (Scheve and Stasavage, 2010).

¹Each estimate is cumulative: it adds up the war's year-by-year effects at the tax system's post-entry persistence, so it measures how much of a war-active country's top rate the war's impulses account for, a given number of years after entry, relative to the year before entry. Because top rates are persistent, the effect at a given horizon compounds the initial increase rather than capturing only that year's change; Section 4 defines this cumulative object precisely.

²We define "war-active" countries as those that sustained military mobilization of at least two percent of their population during the war, and "non-war-active" countries as those that did not, whether they stayed out of the conflict, declared war without mobilizing on that scale, or were invaded and occupied without mobilizing. The non-war-active group is therefore not a set of pure non-participants but the full set of countries against which the war-active countries are compared. Section 3 sets out the classification.

The war sharply widened the gap in top tax rates between war-active and non-war-active countries, and the increase persisted for decades. The raw data shows a gap before the war of roughly seventeen to twenty-six percentage points, averaging twenty-two, that then widens sharply in the 1940s, stays wide for decades, and narrows only as rates fall in both groups after 1980 (Figure 1).

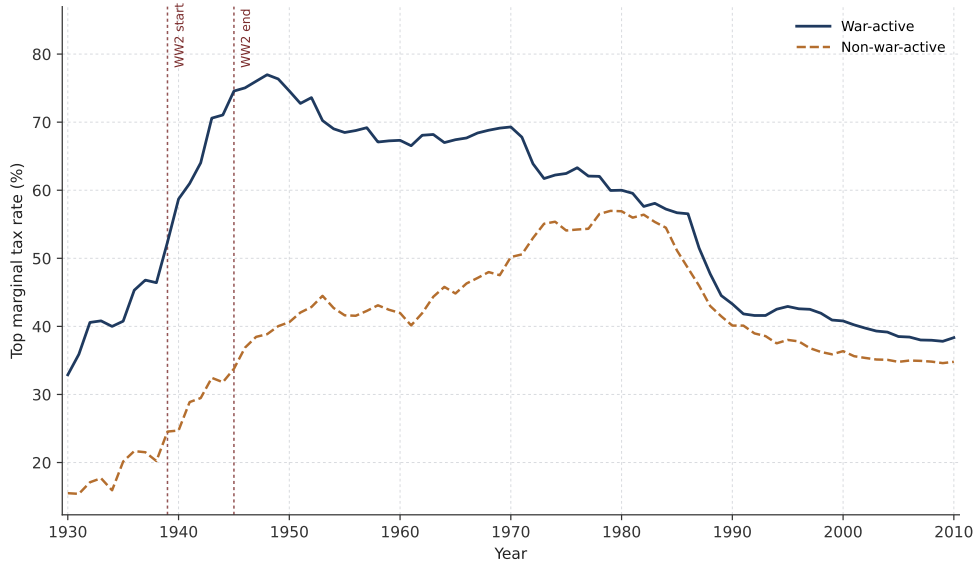


Figure 1: Average top marginal income tax rate by year, 1930–2010, for the 13 war-active and 22 non-war-active countries in the main sample. Each line is the unweighted cross-country mean of the top statutory rate within the group. Vertical lines mark the start and end of the war.

We interpret the persistence of the high top rates as a shift to a new political-economic equilibrium rather than as a by-product of a larger postwar state. The leading alternative is the ratchet effect: war creates permanent revenue needs, so the fiscal state stays large (Higgs, 1987). But a larger state explains the level of postwar revenue, not why the burden kept falling on top incomes. In the account we develop, the war compressed the wealth distribution; the more equal distribution shifted political choices toward more progressive taxation; and that taxation in turn held the distribution compressed, a self-reinforcing loop that kept top taxation high without any further shock and without a change in underlying preferences. We formalize this through the framework of Bénabou (2000).

Among the war-active countries, progressivity rose on three measures beyond the statutory top rate. Across the 7 war-active countries with comparable readings, the average rate the top group actually paid rose from about 25 to 40 percent between the late 1930s and around 1950,

while the statutory top rate in the same countries and years rose from about 53 to 75 percent. The top statutory rate on inheritances, a separate tax on large fortunes, climbs after war entry. The redistribution wedge, the gap between inequality before and after taxes and transfers, is larger in the war-active countries. The three measures point the same way: the war raised progressivity across the tax system, not at the top rate alone.

We contribute to three literatures. Our first contribution is to the literature on warfare, taxation, and fiscal capacity. Its closest precedent for our question is [Scheve and Stasavage \(2010\)](#), who establish that mass mobilization raised the demand for progressive taxation. They neither separate out the Second World War nor trace a war’s effect year by year over the following decades; the two things this paper adds. World War II is an especially suitable case: the conflict sits at the peak of the era in which mass mobilization could raise top taxes ([Onorato et al., 2014](#)). We estimate that war entry raised the top rate by about 32 percentage points within five years and then, to our knowledge for the first time, trace the identified effect of a single war year by year across the following decades, the long-run dimension on which the rest of the paper builds.

Our second contribution is to the literature on long-run institutional persistence. That literature identifies the conditions under which a transitory shock breaks institutional inertia ([Acemoglu and Robinson, 2008](#))³ and documents cases where one has ([Acemoglu et al., 2011](#)), but the shocks are rarely sharp enough, or the outcome observed directly enough, to trace the full trajectory. World War II (among the twentieth century’s largest compressions of inequality, [Scheidel, 2017](#)) and the top rate provide that case: a single event reshaped a core fiscal institution for decades. Applying the same design to World War I yields a similar but smaller effect, a suggestive sign that the pattern is not specific to one war. We also offer an interpretation distinct from the preference-based account of [Scheve and Stasavage \(2010\)](#), in which the war raised the demand for fairness directly: in ours, preferences are unchanged, and a large shock to the wealth distribution moves the economy to a more progressive equilibrium that then sustains itself, which we formalize in Section 5.

Our third contribution is to the literature on cross-country differences in tax progressivity. Much of the theory here is normative: a large literature on optimal taxation asks how taxes ought to be set ([Diamond and Mirrlees, 1971](#); [Judd, 1985](#); [Chamley, 1986](#); [Chari and Kehoe, 1999](#)), but says little about why observed progressivity differs so sharply across countries. Where progressivity

³[Acemoglu and Robinson \(2008\)](#) show that reforms to formal political institutions are offset by countervailing investments in de facto power unless the shock also alters the political structure of society, the identity of the elites, or the source of their economic rents, all three of which mass mobilization changed at once.

is explained at all, it is usually through present-day forces, such as the preferences of the median voter ([Meltzer and Richard, 1981](#)), optimal-tax considerations ([Piketty et al., 2014](#)), or competition for mobile capital ([Keen and Konrad, 2013](#)). Others model it theoretically with limited empirical evidence ([Persson and Tabellini, 2000](#); [Yared, 2010](#)). These explanations ask what sustains progressivity in the present rather than where today’s cross-country differences came from. We offer a historical answer: part of that variation is explained by World War II, not only present-day forces.

The closest precedent for this paper is [Scheve and Stasavage \(2010\)](#), which identifies the effect of mass mobilization on top rates around the First World War but does not separate out the Second. The Second World War enters their analysis only inside a pooled indicator that combines it with the First World War and other conflicts, so the Second World War’s own effect on progressive taxation remains an open question. We identify the Second World War on its own, estimating the effect year by year after each country’s entry, and on a panel of 35 countries, far larger than the samples this literature has relied on: the 8 countries of [Scheve and Stasavage \(2010\)](#) and the 20 countries of [Scheve and Stasavage \(2016\)](#). Tracing the full trajectory over the seven decades that follow is the central result of the paper. That trajectory shows what a pooled design cannot: the Second World War produced not just an immediate rise in top rates but a shift that remained statistically distinct from zero for about four decades, with point estimates staying positive while declining thereafter as top rates fell across countries after 1980. [Saez and Zucman \(2019\)](#) describe that fall as a deliberate policy turn rather than a forced response to globalization; [Frizell \(2025\)](#), analyzing conflict-affected countries from 1960 to 2020, likewise finds that the link between war and progressive taxation itself disappeared after the end of the Cold War. The war’s effect on progressive taxation was therefore long-lived rather than a wartime episode, a claim only a full-path estimate can support.

The paper proceeds as follows. Section 2 sets out the historical background: the wartime fiscal shock, the mass mobilization behind the demand for progressive taxation, the wartime rise in top rates and the compression of wealth, the postwar entrenchment of the increases, and the First World War precedent. Section 3 describes the data: the new 35-country panel of top marginal rates, the mobilization-based country classification, and two complementary progressivity measures. Section 4 presents the event-study strategy and the main result, subjects it to four layers of robustness, and compares it with [Scheve and Stasavage \(2010\)](#). Section 5 develops an equilibrium model in the framework of [Bénabou \(2000\)](#) to interpret the persistence. Section 6 concludes.

2 Historical Background

2.1 The fiscal shock and its financing

The Second World War drove public spending in the belligerent states to levels without peacetime precedent. The war placed every belligerent under fiscal demands (for armies, equipment, and the industrial effort behind them) on a scale their pre-war tax systems had never been designed to raise. Governments were therefore pushed toward extraordinary measures: not marginal adjustments to existing taxes, but a structural expansion of the state's capacity to extract revenue.

Where that revenue was found shaped the fiscal institutions the war left behind. Financing a war by taxing the domestic population, rather than by borrowing abroad, is what builds a state's capacity to assess and collect revenue ([Queralt, 2019](#)). The belligerents of the Second World War met the bulk of the war's cost from domestic resources ([Harrison, ed, 1998](#)), and within that effort the progressive income tax was the lever capable of raising large sums quickly from those most able to pay, alongside domestic borrowing.

2.2 Mass mobilization and the demand for progressive taxation

Mass mobilization generated political pressure to tax the rich, distinct from the war's fiscal need for revenue. By conscripting ordinary citizens into the war effort on a mass scale, the war produced a widely shared demand that owners of wealth bear a comparable burden at home ([Scheve and Stasavage, 2010](#)), and heavier progressive taxation was the form that demand took. What drove this demand was the breadth of mobilization, the share of the population drawn into military and war service, rather than the intensity of combat, so it arose even in countries that saw little fighting on their own soil.

In most belligerents, rearmament through the late 1930s was financed largely by borrowing rather than by broad increases in the taxation of top incomes, most clearly in Germany ([Tooze, 2006](#)). The large, structural increases in top rates came with the mass mobilization of 1939–1941 ([Scheve and Stasavage, 2016](#)), whose fiscal demands far exceeded those of rearmament.

Whether and when a country was drawn into the war was settled largely by events outside its own control. British policy through the late 1930s sought to preserve peace, and the guarantee to Poland of March 1939 was meant to deter a war rather than to wage one.⁴ Deterrence failed:

⁴The clearest expression of this approach was the personal diplomacy of September 1938, when Chamberlain trav-

the invasion of Poland drew Britain and France in within days, and the attack on Pearl Harbor in December 1941 forced the United States to enter ([Weinberg, 1994](#)). Even the war's initiators misjudged its course, since in 1938 neither Germany nor its opponents foresaw the scale the conflict would reach ([Tooze, 2006](#)).

2.3 The wartime rise in top rates and the compression of wealth

The war carried top rates to heights no peacetime system had approached. In the data of Section 3, the war years hold some of the highest rates ever recorded in the panel: the United States reached 94 percent, the United Kingdom 97.5, Canada 95, and the cross-country mean rose from 27 percentage points before the war to 43 during it. These were not marginal adjustments but a rebuilding of the top of the schedule, enacted within a few budget years by governments of every political stripe.

The same years compressed the wealth distribution beneath the schedule. Across the belligerents the war destroyed, expropriated, and devalued large fortunes: physical destruction and occupation extraction where the fighting reached, capital levies at liberation, and everywhere the collapse of asset values and wartime inflation eroding financial wealth ([Scheidel, 2017](#)). In France the top of the wealth distribution lost most of its pre-war holdings ([Piketty, 2001](#)); in Japan the concentration of income at the top collapsed during the war, and that of wealth through the occupation as well ([Moriguchi and Saez, 2008](#)).

The compression did not require destruction at home. In the belligerents the fighting never reached, such as the United States and Canada, top income and wealth shares fell steeply all the same, through the collapse and regulation of asset returns, wartime controls, and the confiscatory taxation of capital income ([Scheidel, 2017](#); [Saez and Veall, 2007](#)).

The two changes arrived together and reinforced each other: the wartime schedule taxed away top incomes while the war itself shrank the fortunes behind them. The postwar political economy inherited both, a steepened schedule and a compressed distribution.

2.4 The postwar welfare state and its persistence

The wartime increase in progressive taxation outlasted the war because the postwar welfare state was built on it. The war reshaped social policy ([Titmuss, 1950](#)) so durably that the wartime expansion

elled to Germany three times to negotiate directly with Hitler over Czechoslovakia, concluding the Munich Agreement. That settlement collapsed when Germany seized the remainder of Czechoslovakia in March 1939 ([Weinberg, 1994](#)).

of public provision created a postwar coalition unwilling to let the new fiscal commitments lapse. Across Western democracies the legislative consolidation of the postwar welfare state clustered in the years immediately after the war (Obinger and Schmitt, 2020); the elevated, progressive taxation built up during the war financed that welfare expansion and was entrenched along with it rather than rolled back at the war's end. That the postwar welfare state embedded these increases, however, does not by itself explain why they endured for decades rather than eroding once the war receded.

We read the durability of these fiscal commitments as evidence that the war had moved the underlying political economy, not merely the tax schedule. A transitory shock leaves lasting institutional change only when it alters not just formal institutions but the political structure of society, the identity of the elites, or the source of their economic rents; otherwise, as Acemoglu and Robinson (2008) show, countervailing investments in de facto power offset the reform and leave outcomes unchanged. Mass mobilization did all three at once: it raised the political weight of ordinary citizens, eroded the prewar concentration of wealth in several belligerents, and narrowed the resources available to those who would resist redistribution.

The elevated rates moved remarkably little once the war ended: as Scheve and Stasavage (2016) document, across the major belligerents they were sustained for three decades through repeated changes of government and shifts in the ideological climate, not only in the immediate postwar years. That durability is more than the breaking of an old equilibrium can account for on its own. Bénabou (2000) supplies a structural reading: inequality and redistribution can support more than one stable configuration, each sustained by the distribution of income and political influence that prevails within it. Once the war had compressed wealth and broadened the political weight of ordinary citizens, the more redistributive of these configurations became self-sustaining, reproducing without the continuing pressure of mobilization. On this reading, the persistence is a property of the equilibrium the war selected, not of any lasting change in preferences; Section 5 formalizes this interpretation.

This equilibrium concerns the progressivity of taxation, which is a separate matter from the size of the fiscal state. That the fiscal state stayed large has a familiar explanation: war pushes public expenditure to a permanently higher level (reconstruction, veterans' pensions, war-debt service) and revenue rises to match (Peacock and Wiseman, 1961; Higgs, 1987). But a higher revenue requirement can be met by taxing broadly rather than progressively, so this ratchet accounts for the size of the postwar fiscal state, not the progressivity of the taxes that financed it.

2.5 Precedent: the First World War

The persistence this paper identifies extends beyond the Second World War. If mass mobilization shifts the political economy toward sustained progressive taxation, the same should be visible after the First World War, the other mass-mobilization conflict of the century. It is: the earlier war produced a similar rise in progressive taxation that likewise outlasted the fighting.

The First World War was a similar shock for the same reason the Second was: a war of mass mobilization, financed increasingly through the income tax. Top marginal rates were pushed to levels without precedent, and much of the population was drawn into the income tax for the first time ([Torregrosa-Hetland and Sabaté, 2022](#)). The wartime increases were partly rolled back afterward, but rates and revenues settled well above their pre-war levels rather than returning to them ([Torregrosa-Hetland and Sabaté, 2022](#); [Peacock and Wiseman, 1961](#)).

This paper also examines the First World War directly. Applying the same event-study design to the earlier conflict yields a smaller but direction-consistent increase in top rates, reported in [Section 4](#). Fewer countries have usable tax data for the period, and the interwar years leave only about a decade before the Depression, so the exercise cannot reach the horizons of the main analysis.

2.6 The post-1980 erosion of top rates

The elevated top rates the war produced were eventually pushed back by forces that had nothing to do with the war. From the early 1980s a new policy consensus took hold across the advanced economies, holding that high top rates discouraged effort and investment, and conservative governments led the turn. In the United States the Reagan administration cut the top federal rate from 70 percent in 1980 to 28 percent by 1988 ([Saez and Zucman, 2019](#)); in the United Kingdom the Thatcher government cut the top rate from 83 to 40 percent between 1979 and 1988; and other countries followed. Tax competition between countries reinforced the downward pressure ([Genschel and Schwarz, 2011](#)). This was a deliberate change in policy, not a forced response to globalization ([Saez and Zucman, 2019](#)).

Because these cuts pulled top rates down across war-active and non-war-active countries alike, the post-1980 decline is a later, common force, distinct from the war's own effect; the empirical analysis in [Section 4](#) holds the two apart.

3 Data

3.1 The top marginal rate and the 35-country panel

The outcome studied in this paper is the top marginal rate of the personal income tax. The data comes from the Comparative Income Taxation Database ([Genovese et al., 2016](#)), which records, for each country and year, the highest statutory marginal rate levied by the national government on personal income, and the law that sets it. Where surcharges or special levies form part of the income tax, the recorded rate combines them; net wealth taxes and sub-national taxes are excluded.

The Comparative Income Taxation Database covers 20 countries, the sample that [Scheve and Stasavage \(2016\)](#) and subsequent work have drawn on. This paper extends it into a new panel of 35 countries, adding 15 whose top-rate series we draw from published country studies, national fiscal histories, and the tax statutes themselves. The additional countries widen the range of wartime experience the data contains, and in particular enlarge the set of countries that were not militarily active in the war. That larger and more varied set of control countries is what allows the war’s effect to be estimated more precisely than the original twenty would permit.

Most of these rates come from a published source: a country study in the top-incomes literature, a national fiscal history, an official statistical publication, a published rate chronology, or a professional tax summary of the kind the major accounting firms publish. The rest come from the tax statutes themselves. Each statute-based rate is carried through every year the law was in force: the law setting the rate and the law next amending it are both identified, so the value between them is the statute’s rate, not an interpolation. A country-year is empty when no published table and no documented statute covers it, and the panel is therefore unbalanced: about one country-year in nine is unobserved, most of them outside the war decades. [Appendix A](#) reports the source for every country.

Top rates vary widely across the panel, both between countries at any moment and within countries over time. The cross-country mean of the top rate rises through the war and early postwar decades, from 27 percentage points before the war to 43 during it and 54 in 1946–1970, and then recedes to 49 in the late twentieth century and 37 in the 2000s ([Table 1](#)). Within-era dispersion stays wide throughout: the range between the lowest and highest national top rates exceeds 70 percentage points in every era from 1929 through 2000 and remains above 40 percentage points in the 2000s. The standard deviation across countries within each era is correspondingly large, on the

order of one-quarter to three-quarters of the period mean, so the upward shift in averages does not amount to a convergence in rates.

Table 1: Top marginal rate across the panel, by era

Era	N	Countries	Mean	SD	Min	Max
Pre-war (1929–1938)	333	35	26.70	20.61	0.00	79.00
War (1939–1945)	239	35	42.73	28.90	0.00	97.50
Early postwar (1946–1970)	785	35	53.61	23.09	0.00	97.50
Late twentieth century (1971–2000)	874	34	49.14	15.67	9.50	97.75
Recent (2001–2010)	316	33	36.55	8.68	11.50	55.00
Full panel (1929–2010)	2,547	35	45.42	21.74	0.00	97.75

Notes: N is the number of country-year observations with a non-missing top marginal rate. Countries is the number of distinct countries contributing to the era. Mean, SD, Min, and Max are computed across country-year observations and reported in percentage points.

3.2 Classifying countries by wartime mobilization

The design compares countries that mobilized for the war with those that did not. Following [Scheve and Stasavage \(2010\)](#), whose mechanism ties the demand for progressive taxation to mass mobilization, we code a country as *war-active* if it sustained military mobilization of at least two percent of its population during the war,⁵ and as *non-war-active* otherwise. The criterion is mobilization, not a declaration of war: several countries declared war without mobilizing on this scale, and they are among the controls. In one respect we apply the criterion more consistently than [Scheve and Stasavage \(2010\)](#), who also require interstate-war participation and so treat high-mobilization neutrals such as Sweden and Ireland as non-participants; we code these two as war-active.⁶ Of the 35 countries, 13 are war-active and 22 are controls (Appendix A lists each). Appendix A.1 reports each country’s mobilization rate, the ratio of sustained wartime military personnel to 1939 population ([Bolt and van Zanden, 2025](#)), with the source for every figure; the interstate-war participation that [Scheve and Stasavage \(2010\)](#) additionally require follows the Correlates of War data ([Sarkees and Wayman, 2010](#)).

Two further countries, Austria and Switzerland, cross the mobilization threshold but are kept

⁵“Sustained” means maintained over the country’s years of participation during the war, as against a brief call-up: the mobilization rate is military personnel divided by 1939 population. Belgium, which demobilized after its 1940 capitulation, does not meet the threshold. Appendix A.1 reports the figure, its concept (peak strength or total raised), and its source for every country.

⁶Both cross the threshold on the mechanism’s own terms. Sweden sustained about 320,000 men under arms and Ireland about 138,000, in each case by conscripting ordinary citizens and carrying the fiscal cost of keeping them mobilized. That is the channel through which mass mobilization raises the demand for progressive taxation, and it operates whether or not the country also fought an interstate war.

as controls on data grounds. Austria’s top rate in 1938–1945 is German tax law applied to occupied territory rather than an Austrian fiscal decision; Switzerland’s wartime fiscal effort ran through a new federal levy rather than its top rate, so the recorded series does not capture it. Section 4.4.2 re-classifies Sweden, Ireland and Austria and the result holds.

3.3 Complementary measures and covariates

A further feature of the tax code complements the top income rate: the top statutory rate on large inheritances and estates. In a progressive schedule, a high inheritance-rate ceiling places the heaviest burden on holders of large fortunes, so it captures how heavily the tax system bears on the transfer of accumulated wealth at the top of the distribution. The series is taken from [Scheve and Stasavage \(2012\)](#), whose data covers the top inheritance rate for 19 of the sample countries through 2000; Appendix Table 9 summarizes it across eras. The inheritance rate is a complementary indicator rather than a primary outcome, and its relationship to the top income rate is examined in Section 4.

A second measure looks past the statutory ceiling to the effective rate: the average income tax rate actually paid by the top income group. The statutory top rate is the marginal rate on the highest bracket, while the effective rate depends on the brackets, deductions, and exemptions below it and is lower. For the 7 countries with comparable pre-war and early postwar readings (Canada, Finland, France, Japan, New Zealand, the United Kingdom, and the United States) we assemble the effective rate from national tax statistics and the countries’ top-incomes studies, documented per country and year in Appendix A.3. The top group is defined differently across these countries, so the effective rate is compared within each country over time rather than across them.

The redistribution wedge measures the progressivity of the tax-and-transfer system as a whole: the gap between pre-tax and post-tax inequality. It captures how much taxes and transfers together compress the distribution, so a wider wedge means a more progressive system in this broader sense. The components come from the World Inequality Database ([World Inequality Database, 2026](#)): the top-10% share of national income and the Gini coefficient, the two broadest measures available, each in a pre-tax and a post-tax version, with the wedge defined as the pre-tax value minus the post-tax value (Appendix Table 10). Comparable post-tax series begin only around 1980 for most countries, so the wedge serves as a supplementary, contemporary indicator rather than a primary outcome.

The robustness analysis also draws on two pre-war country characteristics, income and democracy. Income is real GDP per capita in 1938, in logs, from the Maddison Project Database ([Bolt](#)

and van Zanden, 2025); democracy is the 1938 value of the electoral-democracy index of the Varieties of Democracy project (Coppedge et al., 2025), a continuous scale that rises with the extent of contested and inclusive elections. Both are available for all 35 countries. Section 4.4.3 interacts each with time, letting a country’s pre-war income or democracy move its top rate by a different amount in each later period; neither is an outcome.

4 Empirical Analysis

We treat each war-active country’s entry as an event and ask, year by year after entry, how its top rate moved relative to the path implied by the non-war-active countries and by the country’s own pre-event dynamics. The trajectory the design recovers (a sharp rise at entry, a persistently elevated path for several decades, and a fade at the longest horizons) is what this section identifies as the war’s causal effect.

4.1 Cross-country evidence

The raw data already show the pattern the formal design will recover. In war-active countries the average top rate rises sharply around the time of war entry and stays elevated for decades, while in non-war-active countries it drifts at a much lower level until the post-1980 retreat of top taxation (Figure 2, panel b). The same shape is visible for the First World War on a smaller sample (panel a), where the war-active countries’ top rate lifts sharply during the war and stays higher through the interwar period. A gap separates the two groups before each war, averaging twenty-two percentage points before the Second World War and a few percentage points before the First, and it widens sharply at war entry.

4.2 Empirical strategy

We take a country’s war entry as the event and estimate its dynamic effect on the top rate using an event study with a lagged dependent variable.⁷ The specification is

$$\text{toprate}_{i,t} = \sum_{\tau \in \mathcal{T}} \beta_{\tau} \cdot \mathbf{1}[e_{i,t} = \tau] + \rho \cdot \text{toprate}_{i,t-1} + \delta \cdot \text{Post}_{i,t} \cdot \text{toprate}_{i,t-1} + \alpha_i + \varepsilon_{i,t}, \quad (1)$$

⁷War entry is staggered across three cohorts (1939, 1940, and 1941), which can make pooled event-study coefficients mix cohort effects. Forming the effect cohort by cohort with the heterogeneity-robust estimator of Sun and Abraham (2021) returns the same path as pooling the cohorts (Appendix B.3).

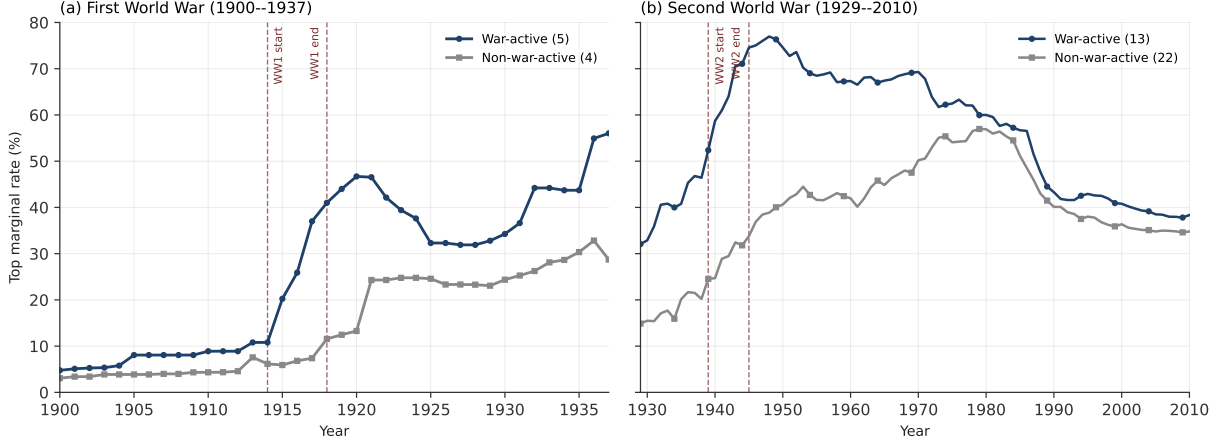


Figure 2: Average top marginal rate by group and year. Panel (a): First World War, on the nine-country sample of war-active countries (ITA, JPN, NZL, GBR, USA) and European-neutral controls (DNK, NLD, NOR, SWE). Panel (b): Second World War, on the full 35-country panel (13 war-active, 22 non-war-active). Each line is the unweighted cross-country mean of the top statutory rate within a group; vertical lines mark the start and end of each war.

where $\text{toprate}_{i,t}$ is the top marginal income tax rate in country i in year t ; $e_{i,t}$ is event time, the year minus country i 's war-entry year; $\mathbf{1}[e_{i,t} = \tau]$ indicates being τ years after entry; β_τ is the per-period effect of war entry on the top rate τ years out; ρ is the persistence of the top rate, common to all countries; $\text{Post}_{i,t}$ is the post-entry treatment indicator, equal to one for a war-active country in the years on or after its war entry and zero otherwise, so it switches from zero to one within a country at entry; δ is the coefficient on the interaction of that indicator with the lagged top rate, $\text{Post}_{i,t} \cdot \text{toprate}_{i,t-1}$, and measures the change in persistence for war-active countries from entry onward; α_i are country fixed effects; and $\varepsilon_{i,t}$ is the residual. The event-time window is $\mathcal{T} = \{-5, \dots, -2\} \cup \{0, \dots, 71\}$, with $\tau = -1$ omitted as the baseline: those observations stay in the sample and only their dummy is dropped. Treated country-years before $\tau = -5$ are excluded from the estimation sample rather than pooled into the baseline, and the sample runs from 1929 to 2010, with the lagged rate for 1929 taken from 1928. An observation enters only if the previous year's rate is available to form the lag. Standard errors come from a wild cluster bootstrap on country, with 999 bootstrap replications drawing Rademacher weights on unrestricted residuals by country cluster (Cameron et al., 2008).⁸

⁸The wild cluster bootstrap is the recommended small-sample correction when the number of clusters is small (Cameron et al., 2008): with only 35 countries (13 treated), standard errors that rely on large-sample normality are unreliable, and the bootstrap, which makes no such distributional assumption, delivers more accurate coverage. Setting the number of replications to $B = 999$ follows standard practice: a bootstrap p -value is computed as $(1 + r)/(B + 1)$, where r counts the replications whose statistic is more extreme than the one observed in the data, so $B + 1 = 1,000$ places conventional significance thresholds exactly on a replication.

Top rates are institutionally sticky: tax systems adjust slowly, and a country’s rate next year is well predicted by its rate this year. Conditioning on the lagged rate does more than absorb this autocorrelation. Last year’s rate already reflects the slow-moving forces that produced it (a country’s fiscal capacity, its political institutions, the prevailing climate of opinion on taxation), so controlling for it nets out the persistent part of those forces, and the post-entry movement is measured relative to where each country’s rate had recently been rather than to its fixed level alone.⁹ The first-order lag is the simplest structure that captures the persistence, and lag-length sensitivity in Section 4.4 shows the persistence estimate and the cumulative trajectory are stable across AR(1)–AR(4). The design follows the precedent of [Acemoglu et al. \(2019\)](#), who use the same lagged-dependent-variable structure to estimate the long-run effect of democracy on GDP per capita, another highly persistent outcome.

The interaction $\text{Post}_{i,t} \cdot \text{toprate}_{i,t-1}$ lets the war change the dynamics of the tax system, not only its level. For a war-active country, persistence is governed by ρ before entry and by $\rho + \delta$ from entry onward, so δ measures the war-induced change in the persistence of the tax institution, not an increment to a fixed level. Without the δ term the specification would force the war to move only the level and leave the adjustment dynamics unchanged, ruling out a shift in how the tax institution itself evolves.

With these components in place, the cumulative effect at horizon $\tau \geq 0$ is the recursive object

$$\Delta_\tau = \beta_\tau + (\rho + \delta) \cdot \Delta_{\tau-1}, \quad \Delta_{-1} \equiv 0, \quad (2)$$

where Δ_τ is the total change in the top rate τ years after entry that the model implies: each year’s new per-period impulse β_τ is added to the carried-forward effect of all earlier years, $(\rho + \delta) \Delta_{\tau-1}$. The boundary condition $\Delta_{-1} \equiv 0$ anchors the path at the entry-baseline year $\tau = -1$, so Δ_τ accumulates the war’s year-by-year impulses at the post-entry persistence: it measures how much of a war-active country’s top rate the war’s impulses account for, relative to the $\tau = -1$ baseline.

The pre-event window is set by the economics of the period. It starts five years before each treated country’s entry, just past the Great Depression (1929–1933, itself a major structural break in tax data).¹⁰

⁹A lagged dependent variable absorbs unobserved determinants of the top rate only insofar as they are carried in the previous year’s rate; it does not control for arbitrary contemporaneous shocks. The alternative specifications in Section 4.4 address those directly, adding year fixed effects and time trends.

¹⁰A longer pre-window would reach back into the Depression and contaminate the pre-period with its tax changes;

Entering the war set in motion the mass mobilization and the political and institutional changes through which top rates rose. We observe few of these changes and cannot measure some of them at all. This is not a gap in the data to be closed: they are consequences of the war, so holding them fixed would remove part of the effect we are measuring. What we estimate is therefore the effect of entering the war together with everything it set in motion, not of the war set apart from its consequences. A causal reading of that effect rests on two conditions about war entry itself, which we set out here and assess with the evidence in Section 4.4: conditional parallel trends and no anticipation.

The first condition, conditional parallel trends, is where the comparability of the two groups enters: the causal reading fails if the countries that mobilized were already moving toward higher top rates for reasons of their own, so that participation and rising progressivity shared a common cause. The condition requires that, absent the war, top rates in the war-active and non-war-active groups would have moved together, once each country’s fixed level and its persistence are accounted for. Country fixed effects absorb fixed cross-country differences in the level of top rates, so a country that taxes top incomes heavily throughout the panel does not by itself drive the estimate. The lagged dependent variable absorbs the persistent component of each country’s path, so a country already drifting upward before entry is not credited with a war effect on that account.¹¹ Broad forces shaped top rates over the whole period, such as the spread of capital mobility and the secular growth of fiscal capacity, but these bore on both groups; the pre-trend test below checks directly that they did not tilt the two groups apart before entry.¹²

The second condition is no anticipation: top rates did not adjust ahead of war entry. The historical record supports this on two counts, laid out in Section 2: in most belligerents, rearmament through the late 1930s was financed largely by borrowing, and the broad increases in top rates came with mobilization; and the timing and scale of mobilization were settled by events outside each country’s control, which even the war’s initiators misjudged. A pre-entry adjustment would have required tax authorities to foresee both the war and its fiscal magnitude, which the events of 1938–1941 gave no government reason to expect.¹³

a shorter one would leave too little pre-event variation to anchor on. Alternative window lengths leave the estimates essentially unchanged: across pre-entry windows of three to ten years, the thirty-year effect lies between +31.7 and +33.4 percentage points.

¹¹This conditioning operates within the estimation and is distinct from the normalization $\Delta_{-1} \equiv 0$ in equation (2), which anchors the cumulative object at the baseline year and does not restrict any pre-entry path.

¹²What distinguishes the war-active group is mass mobilization, the source of variation that the historical account in Section 2 describes.

¹³Other studies likewise treat the Second World War as an exogenous shock for identifying postwar outcomes,

A few changes that coincide with entry might be read as separate causes rather than as part of the war’s effect. An omitted determinant of top rates would bias the estimate only if it were correlated with the timing of war entry and had not itself followed from participation; a determinant unrelated to participation enlarges the residual variance without biasing the effect. The changes that do coincide with entry, postwar reconstruction, Cold-War alliance and military commitments, and the postwar growth boom, are consequences of participation and therefore part of the effect: to the extent they follow from entry, they are part of what the event-time coefficients measure.¹⁴ The Cold-War channel in particular is addressed directly in Section 4.4 by restricting the sample to data through 1970, decades before late-Cold-War military burdens peaked, and the results hold at that cutoff.

A heterogeneity-robust estimator for staggered entry and a sensitivity analysis for parallel-trends violations point the same way and are reported in Appendices B.3 and B.4, respectively.

4.3 Results

The cumulative effect rises sharply at entry, stays large for several decades, and becomes less statistically significant at the longest horizons (Figure 3; Table 2). Five years after entry the war-active country’s top rate is about 32 percentage points above the path the model implies without the war ($\Delta_5 = +32.2$); a decade out it is about 35 ($\Delta_{10} = +34.8$); three decades out, about 33 ($\Delta_{30} = +32.7$). The cumulative effect remains statistically distinct from zero for about four decades and then loses significance as its confidence interval widens ($\Delta_{50} = +8.9$ and $\Delta_{70} = +2.0$, no longer distinct from zero). Because the cumulative effect is built up recursively, its confidence interval at a given horizon compounds the sampling uncertainty in each per-period coefficient and in the estimated persistence; that uncertainty accumulates as the horizon lengthens, and with only 13 war-active countries the long-horizon intervals are correspondingly wide.

including labor-market and human-capital effects (Ichino and Winter-Ebmer, 2004; Akbulut-Yuksel, 2014), innovation (Gross and Sampat, 2023), and the broader compression of inequality (Scheidel, 2017), all treating the war’s incidence as exogenous.

¹⁴To the extent any such factor operated independently of war participation, the country fixed effects and the lagged dependent variable absorb each country’s persistent level and dynamics, and the alternative specifications in Section 4.4 add common trends and year fixed effects that sweep out variables shared across countries; the point estimates shift, and the entry and five-year effects remain distinct from zero in every case (Appendix Table 11). The postwar growth boom affected both groups and so does not generate a divergence between them. Spending tied to reconstruction or alliance commitments raises revenue needs broadly rather than progressively, so it does not fall on the top rate in particular.

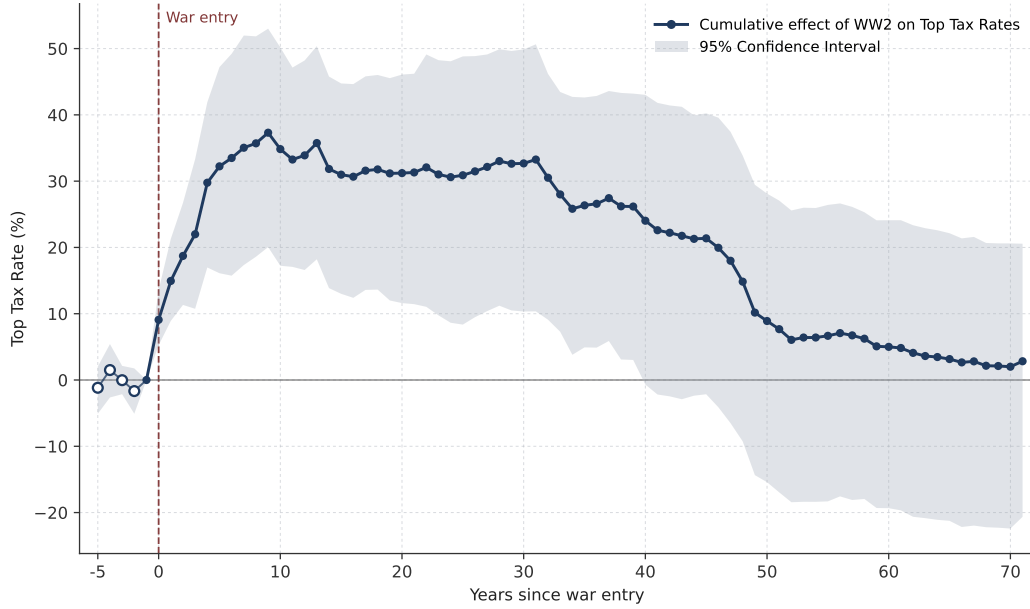


Figure 3: Cumulative effect of WW2 entry on the top marginal rate. The solid line traces Δ_τ over event time; the shaded band is the wild cluster bootstrap 95% confidence interval (999 bootstrap replications). The vertical line at $\tau = 0$ marks war entry; hollow markers at $\tau \in \{-5, \dots, -2\}$ are pre-event placebos.

The pre-event coefficients are economically small and jointly indistinguishable from zero, evidence in favor of the two groups having been on parallel paths before entry.¹⁵ Each individual pre-event per-period coefficient lies within 1.7 percentage points of zero, with a confidence interval that straddles it. In the entry year the per-period coefficient jumps to $\beta_0 = +9.08$ percentage points, already more than four times the largest pre-event coefficient in absolute value: the immediate fiscal demand of the war showing up in top rates the year mobilization begins.

The common persistence of the top rate, estimated across the whole panel, is very high, $\hat{\rho} = 0.94$: this year's rate is almost fully carried into next year's, so top rates change slowly. For war-active countries, entry lowers that persistence slightly, $\hat{\delta} = -0.05$, leaving post-entry persistence at $\hat{\rho} + \hat{\delta} = 0.89$, still high but below the common value. That the war reduced persistence rather than raising it follows from how the war moved tax policy: rates that ordinarily changed very little adjusted sharply during the war, as governments raised them to meet wartime fiscal demands and

¹⁵A joint test of the four placebo leads $\tau \in \{-5, -4, -3, -2\}$ does not reject the hypothesis that they are jointly zero, returning a centered wild cluster bootstrap p -value of 0.919, far above conventional rejection thresholds.

lowered them in the postwar settlements.¹⁶ The two parameters together describe an institutional change rather than a change in level alone: the war raised the top rate through the entry-year jump and also altered the dynamics by which the rate evolves. That reduced persistence concerns the speed of adjustment, not the durability of the elevated level: the sharp wartime adjustments lowered short-run stickiness, while the persistence of the high level is consistent with the political-economic feedback of the new equilibrium rather than with mechanical inertia alone. Because post-entry persistence remains close to one, the entry-year jump is not eroded quickly but carried forward into a path that stays elevated for decades.

The cumulative trajectory grows over the first decade ($\Delta_5 = +32.2$, $\Delta_{10} = +34.8$ percentage points) because the immediate β_0 feeds forward through the high estimated persistence and the early post-event β_τ coefficients add to it (Table 2). The trajectory then plateaus at roughly $\Delta_\tau \approx +32$ percentage points through the first three decades and begins to decay after three decades ($\Delta_{40} = +24.0$, $\Delta_{50} = +8.9$ percentage points). The decay sets in at the horizons that reach into the 1980s and beyond, when top rates fell broadly across countries (Section 2). The war set top rates on a higher path, while the post-1980 decline is a separate, common force, one that the year-fixed-effects specification in Section 4.4 absorbs directly, by removing movements common to all countries within a year, and still leaves a positive effect at three decades.

4.4 Robustness

This section applies four kinds of checks: alternative inference methods; alternative specifications, estimators, samples, and country classifications; pre-war differences between the groups; and progressivity measures beyond the top rate. The cumulative effect remains positive at three decades under every specification that estimates it, and distinct from zero under every inference method and country classification. The specifications that attenuate it, a quadratic trend and year fixed effects, still leave a positive three-decade effect.

4.4.1 Alternative inference methods

With few clusters, the assumed error structure and the asymptotic approximation carry great weight, so we report the cumulative effect under seven inference methods: the wild cluster bootstrap,

¹⁶When re-estimated on 1950–1980 alone, where every war-active country is at least nine years past entry, the persistence shift is $\hat{\delta} = -0.030$: smaller than the full-sample -0.053 but of the same sign and well within the smaller sample’s interval, which includes zero. The persistence shift is not confined to the first post-entry decade.

Table 2: Cumulative effect Δ_τ at standard horizons

τ	Δ_τ (pp)	95% CI low	95% CI high
0	+9.08	+5.24	+13.56
5	+32.23	+16.11	+47.21
10	+34.83	+17.23	+50.18
15	+30.98	+13.00	+44.74
20	+31.21	+11.60	+46.08
30	+32.68	+10.33	+49.87
40	+24.03	-0.69	+43.02
50	+8.90	-15.42	+28.12
60	+5.00	-19.34	+24.10
70	+2.01	-22.41	+20.61

Notes: Recursive AR(1) cumulative $\Delta_\tau = \beta_\tau + (\rho + \delta)\Delta_{\tau-1}$, $\Delta_{-1} \equiv 0$, from the main specification (equation 1): 35-country sample (13 treated, 22 controls), $K = 5$ pre-window, $N = 2,437$. Confidence intervals are the 2.5th and 97.5th percentiles of the wild cluster bootstrap on country (999 replications, Rademacher weights on unrestricted residuals). In each replication the pseudo-outcome is the fitted value plus the sign-flipped residual, with the regressors, including the lagged rate, held at their observed values; the recursion is re-run on each replication, so ρ and δ vary across replications.

standard errors clustered by country, by year, and two-way, heteroskedasticity-robust OLS, Driscoll–Kraay, and the wild block bootstrap (Figure 4; Table 3). At three decades, the effect is significant under all seven. Significance ends after four decades: under the wild cluster bootstrap the interval first includes zero at $\tau = 40$; the ordinary least squares, country-clustered, and wild block bootstrap intervals at $\tau = 41$; the two-way-clustered at 42, the year-clustered at 44, and Driscoll–Kraay at 46; from Δ_{50} on no method excludes zero.

Table 3: Inference comparison: 95% confidence intervals under seven SE methods

Method	Δ_{30} 95% CI	Δ_{50} 95% CI	Δ_{60} 95% CI	Δ_{70} 95% CI
OLS robust (HC1)	[+9.60, +55.75] *	[-14.65, +32.45]	[-18.31, +28.32]	[-21.24, +25.25]
Country-clustered	[+11.72, +53.64] *	[-14.31, +32.11]	[-17.71, +27.71]	[-20.71, +24.72]
Year-clustered	[+11.22, +54.14] *	[-12.78, +30.57]	[-16.59, +26.60]	[-19.61, +23.62]
Two-way clustered	[+12.88, +52.48] *	[-13.00, +30.80]	[-16.53, +26.54]	[-19.62, +23.63]
Driscoll–Kraay	[+12.11, +53.24] *	[-12.08, +29.88]	[-15.87, +25.88]	[-18.84, +22.85]
Wild cluster bootstrap	[+10.33, +49.87] *	[-15.42, +28.12]	[-19.34, +24.10]	[-22.41, +20.61]
Wild block bootstrap	[+9.66, +55.70] *	[-15.00, +32.80]	[-19.09, +29.10]	[-22.16, +26.17]

Notes: The point estimate is the same under every method ($\Delta_{30} = +32.68$, $\Delta_{50} = +8.90$, $\Delta_{60} = +5.00$, $\Delta_{70} = +2.01$ percentage points), so only the 95% confidence intervals are reported. Intervals excluding zero (significance at the 5% level) are marked *. The OLS row uses heteroskedasticity-robust (HC1) standard errors.

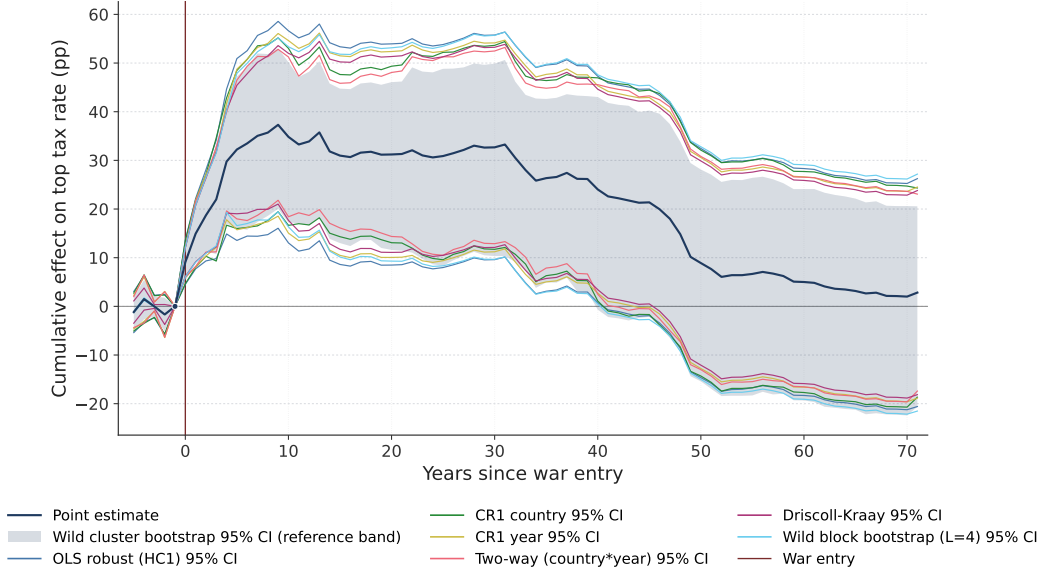


Figure 4: Cumulative effect Δ_τ with 95% confidence intervals under seven inference methods. The point estimate is the same for every method; the bands show how the interval width varies across them.

4.4.2 Alternative specifications, estimators, and classifications

Controlling directly for common movements in top rates leaves the thirty-year effect between +11.8 and +39.6 percentage points. The controls absorb forces shared across countries, such as the post-1980 retreat of top taxation or Cold-War fiscal pressures: a linear trend, a quadratic trend, a post-1980 shift,¹⁷ and year fixed effects, each added to equation (1). The five-year effect is tighter, between +21.8 and +34.0 and distinct from zero in every specification, so the controls matter mainly at long horizons; the main estimate is the median of the five at both horizons, and all five specifications pass the joint pre-trend test (Appendix Table 11). Year fixed effects give the lower bound, since war entry is concentrated in three cohorts and the year dummies absorb part of the war effect itself; under that control the entry effect is +6.6 [+2.3, +11.4] and persistence is unchanged at $\rho + \delta = 0.887$ [0.861, 0.913].

Differential adjustment offers a second alternative: if war-active countries simply adjusted their top rates differently from the controls, most plausibly by cutting more slowly once rates fell everywhere after 1980, the estimated persistence would partly reflect those dynamics rather than the war. Three facts point the other way. Truncating the sample to data through 1970, decades

¹⁷The fall in top rates after about 1980 is closer to a shift in level than to a smooth drift, so a post-1980 indicator captures it with one parameter.

before the post-1980 turn in tax policy, yields the largest estimate of all, $\Delta_{30} = +46.8$ against the full-sample $+32.7$; cuts at 1980, 1990, and 2000 give $+45.3$, $+35.7$, and $+32.1$, respectively. The per-period coefficients at long horizons are negative in the pooled two-way fixed-effects event study without the lagged dependent variable ($\hat{\beta}_{40} = -17.9$, $\hat{\beta}_{50} = -15.9$; Appendix B.3), so in the post-1980 decades war-active countries were cutting faster, not more slowly, with the elevated level carried by persistence. And the raw between-group gap turns negative at long horizons, the opposite of what slower cutting would produce.

Re-estimating with two, three, or four lags of the top rate yields Δ_{30} of $+32.7$, $+32.5$, and $+31.1$ percentage points, against the AR(1) main estimate $+32.7$, with the estimated persistence stable across lag orders at $\rho + \delta = 0.88$ – 0.89 .

No single country drives the result. Dropping each of the 35 countries in turn leaves the trajectory close to the main estimate, with Δ_{30} ranging over $[+25.3, +38.5]$ percentage points around a median of $+32.7$ (Table 12). A model-free difference in group means at each horizon, with no dynamic structure imposed, confirms the early effect: the raw between-group gap is $+17.9$ percentage points at $\tau = 5$ and $+14.1$ at $\tau = 10$, both significant, before fading and turning negative at long horizons as top rates fall across both groups after 1980.

Estimating the effect cohort by cohort returns the same path as pooling the three entry cohorts. Because a country that entered in 1939 would otherwise be available as a comparison for one that entered in 1941, we apply the estimator of Sun and Abraham (2021), which forms each cohort’s effect against never-treated countries only; the two paths differ by less than one percentage point at 69 of the 72 post-entry horizons (Appendix B.3). The HonestDiD analysis of Rambachan and Roth (2023) asks how large a violation of parallel trends would have to be to overturn each coefficient, measured against the largest deviation observed in the pre-period; overturning the entry-year effect would require a violation larger than any seen before the war (Appendix B.4).

The country classification is also not what drives the result. The borderline cases of Section 3.2 are Sweden and Ireland, which the consistent mobilization rule codes war-active, and Austria, which is kept as a control. A strict variant returns Sweden and Ireland to the control group (the classification used by Scheve and Stasavage, 2010, since neither fought an interstate war), an inclusive variant instead adds Austria to the war-active group, counting its Reich-administered wartime rate as a war-active observation, and a third variant does both. All three re-estimate the full event study on the reclassified sample and leave the trajectory close to the main estimate and

statistically distinct from zero at three decades; the strict variant, for instance, gives $\Delta_5 = +35.0$ and $\Delta_{30} = +36.8$, against the main $+32.2$ and $+32.7$ (Appendix Table 13).¹⁸ A finer breakdown by wartime-exposure category, which asks whether the effect extends beyond the active combatants, is reported in Appendix B.5.

4.4.3 Pre-war differences between the groups

The war-active countries were richer and more democratic before the war, and those differences could explain part of the rise in top rates. They lead the rest by about a standard deviation in 1938 income and by two-thirds of one on the electoral-democracy index. If richer or more democratic countries were already moving toward higher top rates, part of what the design attributes to war entry would belong to those characteristics instead.

Allowing pre-war income and democracy to shape the postwar path barely changes the war effect. We interact each country’s 1938 income and democracy with time, so that a richer or more democratic country may follow a different postwar course and any part of that course which follows from those characteristics is absorbed. The thirty-year effect is then between $+29.7$ and $+31.2$ percentage points, against $+32.7$ in the baseline, and the pre-event coefficients remain jointly indistinguishable from zero (Table 4).

Comparing the war-active countries only with others at a similar level of development gives the same answer. This changes the comparison group rather than the specification, keeping only the controls as rich in 1938 as the poorest war-active country, so the two groups are matched on development rather than adjusted for it. The thirty-year effect is $+33.7$ percentage points $[+7.7, +50.3]$. The pre-event coefficients remain jointly indistinguishable from zero.

¹⁸The result also holds if we remove First World War participants from the controls: dropping the nine controls we classify as having any First World War belligerent tie gives $\Delta_{30} = +32.3$; dropping only the four that fought in their own right gives $+31.9$.

Table 4: The effect with pre-war differences allowed their own postwar paths

Specification	Δ_5	Δ_{30}	95% CI	Pre-trend p
Baseline (35 countries)	+32.2	+32.7	[+10.3, +49.9]	0.919
+ GDP $\times$ decade	+31.7	+30.5	[+1.5, +50.0]	0.910
+ GDP & democracy $\times$ decade	+31.5	+29.7	[+0.7, +48.6]	0.905
+ GDP & democracy $\times$ era bands	+31.5	+31.2	[+3.6, +48.8]	0.915

Notes: Cumulative effect on the top marginal income tax rate, five and thirty years after war entry, in percentage points. Each row below the baseline lets a 1938 characteristic follow its own postwar path by interacting it with time; the baseline is the case of no interaction. GDP is log real GDP per capita in 1938 (Bolt and van Zanden, 2025); democracy is the V-Dem electoral-democracy index (Coppedge et al., 2025). “Decade” interacts the characteristic with a dummy for each postwar decade, “era bands” with post-1945 and post-1980 only. Intervals are wild-cluster-bootstrap on country; the pre-trend p -value is from the centered wild-cluster-bootstrap joint test that the pre-event coefficients ($\tau = -5$ to -2) are zero.

4.4.4 Progressivity beyond the top income rate

The war raised progressivity on measures other than the statutory top rate. Three move with it: the effective rate top earners actually paid, the top statutory rate on inheritances, and the redistribution wedge, the gap between pre-tax and post-tax inequality; the latter two are reported in Table 5.

The effective rate rose with the statutory rate. The statutory rate is the marginal rate on the highest bracket, and the effective rate a top taxpayer pays falls below it by the brackets, deductions, and exemptions beneath; in the postwar United States the top statutory rate stood near ninety percent while the top 0.1 percent paid about forty percent on average. Across the 7 countries with comparable readings, the effective rate rose from 24.5 to 39.7 percent between the late 1930s and around 1950. The statutory top rate in the same countries and years rose from 53.4 to 74.9 percent. The effective rate rose by fewer percentage points than the statutory rate but by a larger proportion, so the wartime increase reached real tax payments, not only the schedule.

The top inheritance rate rose with the top income rate. Re-estimating the main specification with the top inheritance rate as the dependent variable, on the 19 countries for which it is available, gives $\Delta_5 = +10.3$, $\Delta_{15} = +19.3$, and $\Delta_{30} = +13.7$ percentage points. A second tax aimed at large fortunes thus moved with the top income rate, though by less, indicating the war steepened the schedule broadly rather than at a single margin.

If the war durably raised the progressivity of the whole system, war-active countries should still operate more progressive systems decades later, even after the gap in top rates between the two groups has mostly closed. The wedge lets us check this implication: it measures how much taxes and transfers together compress the income distribution, so it captures the progressivity of

the system as a whole, of which the top statutory rate is a single parameter. Comparable post-tax series begin only around 1980, so the check is a comparison of levels in the decades where the data exist. The implication holds: the Gini-based wedge is 3.9 points larger in the war-active group [+1.1, +6.8] and the top-10% wedge 2.3 points larger [+0.6, +4.0].

There is no tension between this gap and the narrowing of top rates over the same decades. The top rate is one parameter that a single reform can move, and the post-1980 cuts closed most of the gap in top rates between the two groups. The system behind the wedge is an accumulated structure of schedules, social insurance, and transfers, built up in the postwar decades, that no single reform can undo (Lindert, 2004), as Section 2 described. The divergence the war created therefore closes first at the top rate, the parameter a reform can move at once, and lingers in the slower-moving system. The gap is a comparison of levels rather than an identified effect, but it is the pattern the persistence account predicts.

Table 5: Progressivity beyond the top income rate

<i>Panel A. Tax schedule: cumulative effect after war entry</i>				
Outcome	Sample	Δ_5	Δ_{15}	Δ_{30}
Top marginal income rate (main estimate)	35 countries	+32.2	+31.0	+32.7
Top inheritance rate	19 countries	+10.3	+19.3	+13.7
<i>Panel B. Redistribution wedge: war-active vs non-war-active difference, post-1980</i>				
Outcome	Sample	Difference	[95% CI]	
Gini-based wedge	34 countries	+3.9	[+1.1, +6.8]	
Top-10% wedge	34 countries	+2.3	[+0.6, +4.0]	

Notes: All figures are percentage points. Panel A reports the cumulative effect Δ_τ five, fifteen, and thirty years after a country's war entry. Panel B reports the difference in the redistribution wedge, the gap between pre-tax and post-tax inequality, between the war-active and non-war-active groups over 1980–2010, the part of the panel for which post-tax series exist. Confidence intervals are country-clustered (CR1) at 95 percent.

4.5 Comparison with Scheve and Stasavage (2010)

Comparing our results with Scheve and Stasavage (2010) shows what the single-war design adds. Their mobilization indicator pools the two World Wars, the Franco-Prussian War, and the Korean War into one coefficient, across 8 countries in a panel that ends in 1970, so it answers whether mass mobilization in general raised top rates. Two findings here could not be recovered from that design. First, the Second World War has its own identified effect: about 32 percentage points within five years. Second, that effect has a full trajectory, a plateau near 33 points for three decades and a decline after 1980, which a panel ending in 1970 cannot capture.

Where the designs overlap, on the First World War, our estimate is somewhat smaller: +27.0 percentage points five years after entry, against the roughly 31 to 36 percentage points they report for their belligerents; it eases to +19.6 by ten years (Appendix A.2). The mechanism we propose differs: Section 5 formalizes an equilibrium account in which the war changes the wealth distribution rather than preferences.

5 A Model of War and Progressive Taxation

This section offers an economic interpretation of the empirical findings, building on the multiple-equilibria framework of [Bénabou \(2000\)](#) to show why a transitory shock can have lasting effects. The estimates do not test the model. The question it answers is why a one-time shock to inequality, the war, can produce changes in top tax rates that persist for decades without any change in preferences. The answer is multiple equilibria: a sufficiently large shock moves the economy between basins of attraction, and the equilibrium it selects then sustains itself under unchanged primitives.¹⁹

5.1 The mechanism and its relationship to Bénabou (2000)

We build on the multiple-equilibria framework of [Bénabou \(2000\)](#), in which the redistribution a society chooses and the inequality it lives with are determined jointly, so that the same fundamentals can support more than one stable outcome. The logic is intuitive: a more equal distribution of wealth strengthens political support for redistribution, and more redistribution in turn keeps the distribution equal, so a large enough shock to inequality can move the economy from a low-redistribution outcome to a high-redistribution one that then sustains itself. We make two adaptations to [Bénabou \(2000\)](#). We simplify preferences to log utility, so the case for redistribution does not rest on a high coefficient of relative risk aversion;²⁰ and we bring the framework to bear on an identified shock, the Second World War, that Bénabou never models.

¹⁹Several symbols in this section carry model-specific meanings distinct from the empirical sections: τ is the tax-progressivity parameter (not event time), ρ the bequest rate (not the persistence of the top rate), δ the labor share in production (not the war-induced persistence shift of equation 1), and α the capital share (not a country fixed effect α_i).

²⁰In [Bénabou \(2000\)](#), the strength of the redistribution motive rises with the curvature of utility: the more risk-averse agents are, the more they value the insurance that progressive taxation provides, so a high coefficient of relative risk aversion makes redistribution easier to sustain. Log utility fixes this coefficient at one, so the multiplicity here does not depend on assuming strong risk aversion.

5.2 The model

The economy is populated by a continuum of dynasties indexed by $i \in [0, 1]$, with overlapping generations. Each adult lives one period, inheriting capital from the parent, supplying labor, producing output, paying taxes, and bequeathing capital to the child.

Preferences are log over consumption c_t^i and the child's capital endowment, with additive disutility of labor effort ℓ_t^i :

$$U_t^i = (1 - \pi) \left[\ln c_t^i - \frac{(\ell_t^i)^\eta}{\eta} \right] + \pi \ln k_{t+1}^i, \quad (3)$$

where $\pi \in (0, 1)$ governs the strength of the bequest motive, $\eta > 1$ controls labor-disutility curvature (with Frisch elasticity $1/(\eta - 1)$), and ℓ_t^i is labor effort. Output is

$$y_t^i = z_t^i (k_t^i)^\alpha (\ell_t^i)^\delta, \quad (4)$$

with capital share α , labor share δ , and an idiosyncratic productivity shock z_t^i with $\ln z_t^i \sim \mathcal{N}(-v^2/2, v^2)$. The cross-sectional variance of log capital, $\sigma_t^2 \equiv \text{Var}(\ln k_t^i)$, is the state variable that summarizes inequality.

Fiscal policy follows Bénabou's residual-progression scheme:

$$\hat{y}_t^i = (y_t^i)^{1-\tau_t} \cdot \tilde{y}_t^{\tau_t}, \quad (5)$$

where $\tau_t \in [0, 1]$ is the progressivity parameter and $\tilde{y}_t$ is set by budget balance, $\int \hat{y}_t^i di = \int y_t^i di$. The variance of log after-tax income is exactly $(1 - \tau_t)^2$ times the variance of log pre-tax income, which makes the redistribution channel transparent: $\tau_t = 0$ is laissez-faire and $\tau_t = 1$ is full equalization.

Capital accumulates through two channels:

$$k_{t+1}^i = (k_t^i)^\theta \cdot (\rho \hat{y}_t^i)^{1-\theta}, \quad \theta \in [0, 1], \quad (6)$$

where $\rho = \pi(1 - \theta)/(1 - \pi\theta)$ is the bequest rate (the log-utility savings rate derived from (3)), and θ is the share of intergenerational transmission that bypasses the tax-and-transfer system. This direct-transmission channel covers the persistent non-financial inheritances (genetic endowments, social networks, professional contacts, family name, cultural capital) that are not income and not taxable. The two-channel form follows the structure of the accumulation equation in [Bénabou](#)

(2000), with θ taking the role of his direct-transmission term. Its substantive role here is to give the inequality dynamic a non-tax-mediated floor, which connects to the wartime destruction of concentrated, hard-to-tax wealth (Section 4.4.4 reports that the war also raises the inheritance top rate, the most direct measure of taxation on intergenerational transfers).

Cross-generation persistence in capital is

$$P(\tau) = \theta + (1 - \theta)\alpha(1 - \tau). \quad (7)$$

Even at $\tau = 1$, $P = \theta > 0$: redistribution does not remove the direct-transmission floor. The cross-sectional variance evolves according to

$$\sigma_{t+1}^2 = P(\tau_t)^2 \sigma_t^2 + (1 - \theta)^2 (1 - \tau_t)^2 v^2, \quad (8)$$

so the steady-state inequality given a fixed τ is

$$\sigma_{ss}^2(\tau) = \frac{(1 - \theta)^2 (1 - \tau)^2 v^2}{1 - P(\tau)^2}, \quad (9)$$

which is strictly decreasing in τ over $[0, 1)$ with $\sigma_{ss}^2(1) = 0$.

Political equilibrium is set by myopic voting: agents vote on τ_t before z_t^i realises and do not internalise the effect of τ_t on future inequality. The pivotal voter is wealth-biased, following Bénabou's ordinal scheme (Bénabou, 2000, Prop. 6),

$$\ln k_t^{\text{piv}} = m_t + \lambda \sigma_t, \quad (10)$$

where $m_t = E[\ln k_t^i]$, $\sigma_t = \sqrt{\sigma_t^2}$, and $\lambda > 0$ captures the disproportionate political influence of wealthier voters. The pivotal voter's preferred tax rate solves the first-order condition

$$(1 - \tau)(\alpha^2 \sigma^2 + v^2) - \alpha \lambda \sigma - \frac{\delta}{\eta} \cdot \frac{\tau}{1 - \tau} = 0, \quad (11)$$

which implicitly defines the political curve $\tau = T(\sigma^2)$. The three terms have a clean reading: $(1 - \tau)(\alpha^2 \sigma^2 + v^2)$ is the marginal benefit of redistribution, combining an insurance motive proportional to v^2 and an inequality-reduction motive proportional to $\alpha^2 \sigma^2$; $\alpha \lambda \sigma$ is the marginal opposition from the wealth-biased pivotal voter; $(\delta/\eta) \cdot \tau/(1 - \tau)$ is the marginal deadweight loss from the labor

distortion. See Appendix C for derivations of the bequest rate, the labor first-order condition, the variance law of motion, and the political first-order condition.

5.3 Multiple equilibria

A steady-state political-economic equilibrium is a pair (σ_{ss}^2, τ^*) that satisfies both $\tau^* = T(\sigma_{ss}^2)$ and $\sigma_{ss}^2 = \sigma_{ss}^2(\tau^*)$: the pivotal voter's preferred tax rate at the inequality level, and the inequality level sustained by the law of motion under that tax rate. Geometrically, an equilibrium is an intersection of the political curve and the inequality locus in (σ^2, τ) space.

Under the illustrative calibration $\theta = 0.7$, $\alpha = 0.95$, $v^2 = 1.0$, $\lambda = 2.0$, $\delta/\eta = 0.01$ the two curves intersect at three points (Figure 5): a high-inequality, low-redistribution steady state at $(\sigma^2, \tau) \approx (1.82, 0.03)$; an unstable threshold at $(0.068, 0.524)$; and a low-inequality, high-redistribution steady state at $(0.014, 0.746)$. The two outer steady states are stable in the sense that small perturbations decay back toward them; the middle one is unstable and separates the basins of attraction of the two stable equilibria. The multiplicity is presented here as a property of the calibrated model that gives the war shock its structural foothold; the exact parametrization and the verified intersection values are in Appendix C.

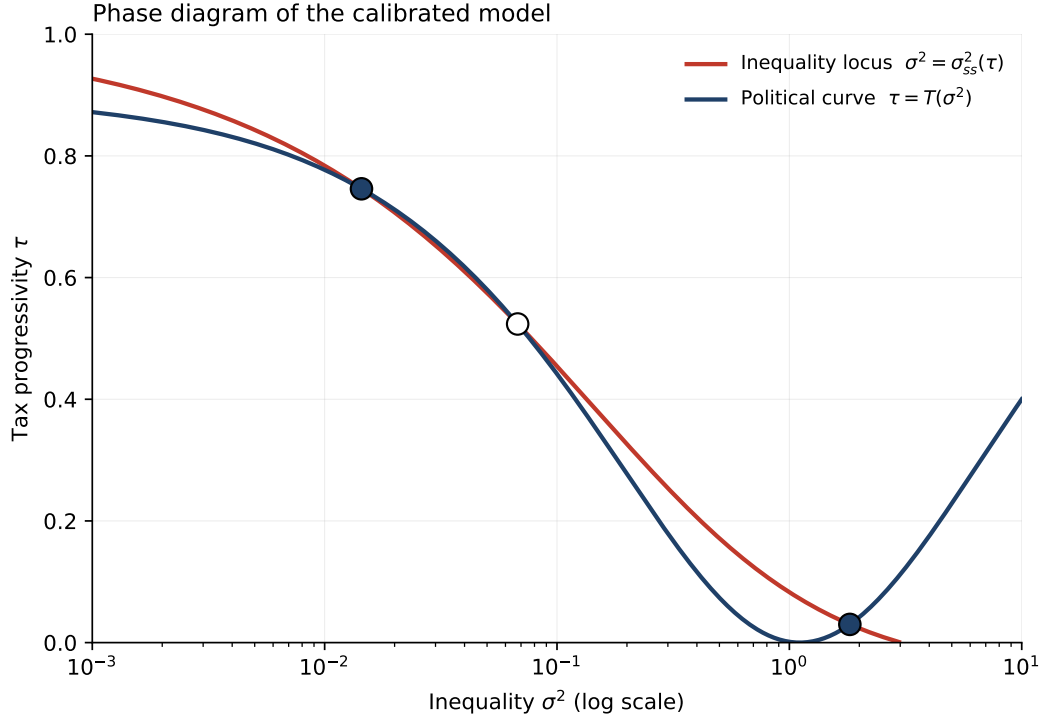


Figure 5: Phase diagram of the calibrated model. The inequality locus $\sigma^2 = \sigma_{ss}^2(\tau)$ (red) and the political curve $\tau = T(\sigma^2)$ (blue) intersect at three points: two stable (filled), and one unstable (open) that separates their basins of attraction. Inequality on a log scale. Calibration $\theta = 0.7$, $\alpha = 0.95$, $v^2 = 1.0$, $\lambda = 2.0$, $\delta/\eta = 0.01$.

5.4 The war as a shock that selects an equilibrium

The war is modeled as a one-time exogenous shock to σ^2 , the cross-sectional variance of log capital. The primitive parameters $(\theta, \alpha, v^2, \lambda, \delta/\eta)$ are unchanged: preferences, the production technology, the wealth-bias mapping from inequality to political influence, and the labor distortion are all unaffected. What changes is the state variable: progressive capital destruction during the war (capital levies, factory destruction, wartime expropriation, the wartime collapse in financial-asset valuations) falls disproportionately on top wealth-holders and compresses the cross-sectional distribution.

Whether the shock has a permanent effect depends on whether it pushes σ^2 below the unstable threshold. If it does not, the economy stays in the basin of the pre-war steady state and inequality and the tax rate drift back to their pre-war values. If it does, the economy enters the basin of the post-war steady state: the political response to the lower σ^2 is an immediate jump in τ , after which the joint dynamics carry σ^2 further down and τ further up toward the post-war steady state. Figure 6 traces a simulated path for a shock that crosses the threshold: τ jumps from the pre-war

value of about 0.03 to about 0.68 in the period of entry, then drifts slowly toward the post-war steady state at about 0.75.

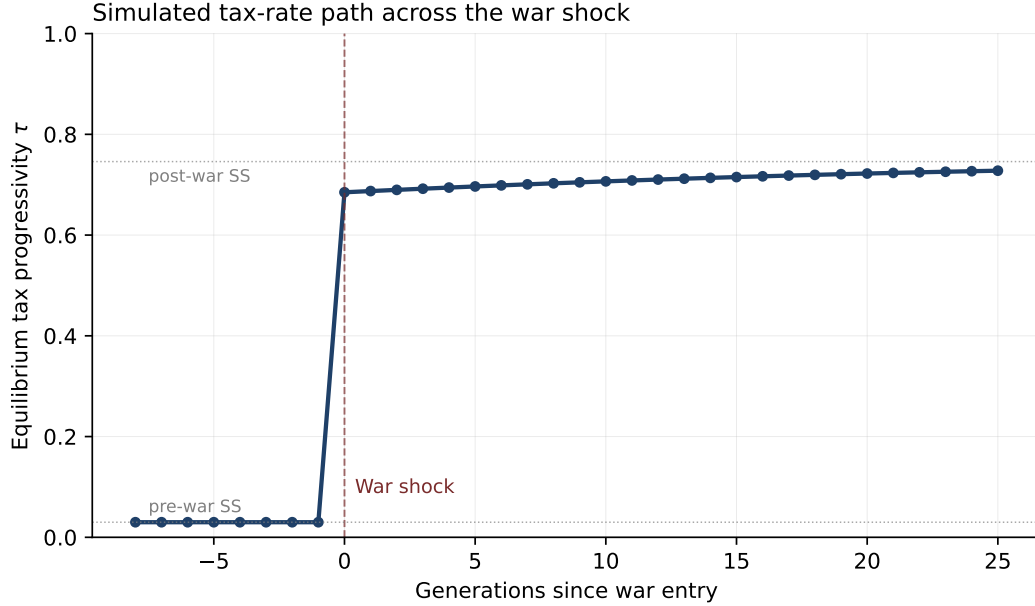


Figure 6: Simulated equilibrium tax-rate path across the war shock. The pre-war steady state $\tau \approx 0.03$ is sustained for several generations; at $t = 0$ a one-time shock displaces σ^2 below the unstable threshold. The political response is immediate (τ jumps to about 0.68), and the joint dynamics carry the economy slowly toward the post-war steady state at $\tau \approx 0.75$.

The contrast with the preference-based account of [Scheve and Stasavage \(2010\)](#) is clear. In their account, the war changed preferences: the experience of mass conscription raised the demand for fairness, and the war-active electorate that emerged after 1945 was different from the one that entered the war. In the account formalized here, preferences are fixed and the war changed the distribution; the same political rule, applied to a more compressed distribution, sustains a higher level of redistribution, which then sustains the distribution.

A change in preferences could in principle select the same equilibrium, but the framework treats the two shocks asymmetrically. Any change in preferences or in the political rule moves the wealth distribution only through the tax system, and the tax system compresses only the part of inherited advantage that passes through taxable income: the direct-transmission share θ lies beyond its reach. From the law of motion (8), even full equalization ($\tau = 1$) reduces the variance of inherited capital by no more than the factor θ^2 each generation. A preference-driven shift is therefore gradual by construction. It must hold redistribution high for decades before the distribution has moved enough to sustain the new equilibrium on its own; if the preference change fades sooner, the tax rate returns

to its pre-war level. A permanent change in preferences does not fade, but it predicts a slow drift in top rates rather than a jump, and to accommodate the post-1980 decline it must be followed by a second change in the opposite direction.

The war operates on the state directly. Capital destruction, expropriation, and the collapse of asset values compress the distribution at once, without passing through the fiscal channel, so the political response is immediate and the new equilibrium sustains itself under unchanged preferences. This asymmetry matches the three features of the evidence: the sharp rise at entry, the persistence across decades without further shocks, and the post-1980 reversal, which reads as a policy turn operating on the equilibrium. The preference account can be made to fit the same facts, but only with more moving parts: the demand for fairness that [Scheve and Stasavage \(2010\)](#) document must itself persist for decades and then reverse, and a reversal around 1980, when the cohorts that lived through the war were leaving the electorate, is not implausible. Our account needs neither the persistence nor the reversal of preferences: a single shock to the distribution, a political rule that never changes, and a later policy turn account for the whole path.

The threshold-crossing structure maps qualitatively to the exposure-category ordering in [Appendix B.5](#). Larger shocks are likelier to cross the threshold and select the high-redistribution basin; smaller shocks are more likely to leave the economy in the pre-war basin. Empirically, active combatant countries received the largest distributional shocks and show the most durable post-event divergence, while marginal participants show smaller and less durable effects. The mapping is qualitative, since the model does not deliver an exact threshold in observable shock magnitudes.

The First World War is a case where the two accounts part ways. It, too, was a war of mass mobilization, so if a change in preferences alone selected the high-redistribution equilibrium, it should have done so after 1918 as much as after 1945. The rise it produced was less durable: top rates rose sharply during the war and were then partly rolled back in the 1920s, both in the raw data ([Section 2](#)) and in the event-study estimate ([Section 4.5](#)), which rises by somewhat less than the Second World War's in the first five years and then declines over the following decade. Our account explains the difference through the size of the shock to the distribution: a compression too small to carry the economies across the threshold produces a rise that is partly rolled back rather than locked in. What separates the two wars, on this reading, is not how far preferences moved but how far wealth did.

6 Conclusion

The Second World War raised top marginal tax rates sharply and durably in the countries that mobilized for it. Five years after entry, a war-active country's top rate was about 32 percentage points higher than the path implied by the non-war-active control group and the country's own pre-event dynamics; three decades out the effect was about 33 percentage points, and it remained statistically distinct from zero for about four decades before fading with the broad fall in top rates after 1980. The estimates hold under alternative inference methods, specifications, and country classifications, and the same design applied to the First World War yields a similar but smaller effect.

We read the persistence as a shift between political-economic equilibria rather than as a change in preferences. By compressing the wealth distribution at the top, the war moved these countries into a basin of attraction with higher redistribution that then sustained itself: with preferences and the rules translating wealth into political influence unchanged, the compressed distribution shifted political choices toward higher top taxation, and the high top taxation kept the distribution compressed. This mechanism is distinct from the fairness-preference account, in which the war changed what voters wanted; here, what changed is the distribution their politics operates on.

Part of the variation in tax progressivity across countries in the second half of the twentieth century is the imprint of this one shock. Differences often read as expressions of stable national preferences are, on this account, the lasting mark of an event that reshaped a core fiscal institution and the distribution it taxes at once. Progressivity then persists as an equilibrium, held in place by the institutions the war built and the compressed distribution they maintain, until a shock large enough to reset it arrives; the post-1980 erosion of top rates was such a shock, a deliberate policy turn operating on the equilibrium the war had selected.

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A Source documentation

Table 6 reports the source of the top marginal rate series for each of the 35 countries, and the group to which the country is assigned in the analysis.

Table 6: Top marginal rate source by country

Country	Source for top-rate series	Group
Australia	Comparative Income Taxation Database (CITD)	T
Austria	CITD	C
Belgium	CITD	C
Canada	CITD	T
Denmark	CITD ^a	C
Finland	CITD	T
France	CITD	T
Germany	CITD	T
Ireland	CITD	T
Italy	CITD	T
Japan	CITD	T
Netherlands	CITD	C
New Zealand	CITD	T
Norway	CITD	C
South Korea	CITD (1949–2010); 1933–1940 from Kim and Kim (2015), Table B5; 1929–1932 zero per their Appendix A.1 (no personal income tax before 1934)	C
Spain	CITD	C
Sweden	CITD	T
Switzerland	CITD	C
United Kingdom	CITD	T
United States	CITD	T
Argentina	Alvaredo (2010), “The Rich in Argentina over the Twentieth Century,” Table 6.3; Argentine tax statutes (Ley 11.682/1932 onward) via InfoLeg and the Diario de Sesiones	C
Brazil	Nóbrega (2014), <i>História do Imposto de Renda no Brasil</i> , Receita Federal do Brasil; Brazilian tax statutes (Lei 4.154/1962, Lei 7.713/1988, Lei 9.250/1995 onward); Receita Federal Memória portal	C
Chile	Chilean tax statutes (DL 755/1925, Ley 5169/1933, Ley 6334/1939, Ley 8419/1946, Ley 9629/1950, DS 2106/1954, Ley 11575/1954, Ley 13305/1959, Ley 16466/1964, DL 824/1974, Ley 18985/1990, Ley 19753/2001) via BCN LeyChile; Biehl, Labarca and Atria (2025); Flores, Sanhueza, Atria and Mayer (2020); SII circulars	C
Colombia	Colombian tax statutes (Ley 64/1927, Ley 78/1935, Ley 45/1942, Decreto 2317/1953 onward) via SUIN-Juriscol; Mora Toscano (2013); Junguito and Rincón (2004)	C
Costa Rica	Facio (1948), “El impuesto sobre la renta y la economía nacional,” <i>El Trimestre Económico</i> ; Ley 837/1946 and Ley 1789/1954 via SCIJ	C
Guatemala	ICEFI (2007), <i>Historia de la Tributación en Guatemala</i>	C
Iceland	Skatturinn (Iceland Revenue and Customs), tax-history annals; Stefánsson, centenary history of the income tax; Lög nr. 74/1921 onward	C
India	Indian Finance Acts 1929–1960 as printed in the official Acts collections; <i>Statistical Abstract of India</i> ; Taxation Enquiry Commission (1953–54); Banerjee and Piketty (2005), Figure 8; budget-rate compilations	C
Mexico	Ibarra Morales (2020), <i>La contribución del Impuesto sobre la Renta a la hacienda pública federal de México</i> , UNAM (rates as published in the Diario Oficial de la Federación)	C
Peru	Castillo-García (2026), <i>Conflicting Claims and Taxation</i> (World Inequality Lab), Figure 11; Ley 7904/1934; Decreto Legislativo 774/1993 and its consolidations	C
Philippines	Philippine statutes (Act 2833, Act 2926, CA 117, CA 466, RA 82, RA 590, RA 1094, RA 2343, EO 37, RA 8424) via the LawPhil Project	C
Portugal	Alvaredo (2010), “Top Incomes and Earnings in Portugal,” Tables 11A.2 and 11C.1; Instituto Mais Liberdade, IRS rate chronology	C
South Africa	Alvaredo and Atkinson (2016), “Top Incomes in South Africa over a Century,” Table A.3A	T
Uruguay	García Repetto (2005), “El impuesto a la renta,” <i>Boletín de Historia Económica</i> ; Ley 12.804/1960 via IMPO	C
Venezuela	Fariñas, “Obligaciones fiscales de la industria petrolera venezolana” (Libro Homenaje, UCV); García (2021), <i>Actualidad Contable Faces</i> ; Ley de Impuesto sobre la Renta de 1966; Gaceta Oficial statute texts; PwC and KPMG country tax summaries	C

Notes: CITD denotes the Comparative Income Taxation Database (Genovese et al., 2016). “Group” classifies each country as war-active (T) or non-war-active (C) for the analysis; the classification is set out in Section 3.2. ^a For Denmark the database records the equilibrium marginal rate of Atkinson and Sogaard in place of the statutory rate, throughout the series. Danish formal rates were set net of the deduction of the tax itself and so ran to more than one hundred percent between 1951 and 1966; the equilibrium rate is the comparable measure. Denmark is a control country and this measure is used only on that side of the comparison.

A.1 Mobilization classification

A country is coded war-active if it sustained military mobilization of at least two percent of its population during the war, the threshold of [Scheve and Stasavage \(2010\)](#). Table 7 reports, for each treated country and for the borderline cases, sustained wartime military strength, 1939 population, and the resulting mobilization rate. The war-active countries range from 3.3 percent (South Africa) to 19.0 percent (Finland). India and Brazil fall well below the threshold and are therefore controls, though the interstate-war component of their rule would count them as participants.

Table 7: Wartime mobilization rates and the war-active classification

Country	Personnel (000s)	Population (000s)	Rate (%)	Source
<i>War-active (treated)</i>				
Australia	993	6,971	14.2	Ellis (1993)
Canada	1,100	11,570	9.5	Ellis (1993)
Finland	700	3,686	19.0	Kinnunen and Kivimäki, eds (2012)
France	5,000	41,900	11.9	Harrison, ed (1998)
Germany	9,480	69,286	13.7	Harrison, ed (1998)
Ireland ^a	138	2,934	4.7	Carroll (1975)
Italy	3,815	43,865	8.7	Harrison, ed (1998)
Japan	7,730	72,364	10.7	Harrison, ed (1998)
New Zealand	140	1,627	8.6	New Zealand Ministry for Culture and Heritage (2012)
South Africa ^b	334	10,170	3.3	Wessels (2000)
Sweden	320	6,326	5.1	Carlgren (1977)
United Kingdom	5,090	47,991	10.6	Harrison, ed (1998)
United States	11,430	131,539	8.7	Harrison, ed (1998)
<i>Borderline, kept as controls</i>				
Austria ^c	—	6,653	> 2	—
Switzerland ^c	650	4,206	15.5	Singer et al. (1972)
India	2,582	381,400	0.7	Ellis (1993)
Brazil	105	40,289	0.3	Singer et al. (1972)

Notes: Personnel is wartime military strength as reported by the cited source (peak annual strength for the Harrison figures; total personnel raised for the Ellis figures); population is 1939 population from the Maddison Project ([Bolt and van Zanden, 2025](#)); both are in thousands, and the rate is personnel divided by population.

^a Ireland's figure includes the part-time Local Defence Force; the regular army alone is about 1.4 percent, so Ireland clears the threshold only on the broader measure. ^b South Africa's 1939 population is not reported by the Maddison Project and is taken from the Correlates of War National Material Capabilities data (v6). War entry years are 1939 for Australia, Canada, Finland, France, Germany, Ireland, New Zealand, South Africa, Sweden and the United Kingdom; 1940 for Italy and Japan; and 1941 for the United States. ^c Austria and Switzerland exceed the threshold but are kept as controls on data-quality grounds (Section 3.2): Austria's wartime top rate is German tax law applied to occupied territory, and Switzerland's wartime fiscal effort ran through a new federal levy rather than its top rate.

A.2 First World War data

The First World War exercise (Figure 2, panel (a), and the estimate in Section 4.5) uses a nine-country sample: five war-active countries (Italy, Japan, New Zealand, the United Kingdom, and

the United States) and four European neutrals (Denmark, the Netherlands, Norway, and Sweden). These are the countries in the Comparative Income Taxation Database whose top-rate series extend far enough before 1914 to support a pre-war baseline. The top marginal rate comes from the same source used for the main panel, the Comparative Income Taxation Database ([Genovese et al., 2016](#)), read back to the early years of each series. The exercise is estimated on 1908–1928 under the specification of equation (1), with the same five-year pre-window and $\tau = -1$ as the sole omitted reference. The window stops in 1928 so that no horizon falls in the Depression, when top rates rose in both groups for reasons unrelated to the war; horizons are estimable to $\tau = 14$. A country is coded war-active if it was a belligerent in the First World War and neutral otherwise, and each war-active country’s entry is dated to the year it entered the war: 1914 for the United Kingdom, Japan, and New Zealand, 1915 for Italy, and 1917 for the United States ([Sarkees and Wayman, 2010](#)). With nine countries and a window ending in 1928, we read these estimates as descriptive.

A.3 Effective-rate sources

Table 8 lists, for each country, the source of the effective (average) income tax rate on the top income group, the income group on which it is defined, and the years covered.

Table 8: Sources for the top-group effective income tax rate, by country

Country	Top group	Source	Years	Obs.
Canada	Top 0.1%	Saez and Veall (2007) , average tax rates in upper income groups	1920–1960	16
Finland	Top 0.1%	Official income and property tax statistics (SVT IV B); tax-unit totals from Jäntti et al. (2010)	1938–1952	4
France	Top 0.1%	Piketty (2001) , statistical annexes (Tableaux B-19/B-20)	1915–1998	16
Japan	Top 0.1% of adults ^a	Statistical Yearbook of the Empire of Japan and Japan Statistical Yearbook; reference totals from Moriguchi and Saez (2008)	1929–1953	2
New Zealand	Top income grade ^a	New Zealand Official Yearbooks, “Incomes and Income-tax”; population totals from Atkinson and Leigh (2007)	1935–1954 ^b	12
United Kingdom	Top 1%	Torregrosa-Hetland and Sabaté (2026) , microsimulation on Inland Revenue distribution tables	1911–1949	2
United States	Top 0.1%	IRS Statistics of Income, actual rates paid	1916–1960	16

Notes: Obs. is the number of years with an effective-rate observation between 1938 and 1953; the comparison in Section 4.4.4 uses each country’s rate for 1938 and for the year nearest 1950. The rate is the average income tax assessed on the top group as a share of that group’s reported income. ^a For Japan the rate is constructed at a fixed 0.1 percent of adults by log-linear interpolation between published income classes, and postwar self-assessed figures exclude wage income taxed by withholding; for New Zealand it is the top observed income grade and excludes the flat Social Security Charge and National Security Tax. ^b No distribution statistics for the income years 1941/42–1944/45 (wartime gap).

B Empirical Analysis: Additional Tables

This appendix collects the additional tables and figures behind Sections 3 and 4, in the order the text refers to them: descriptive statistics for the complementary progressivity measures (Section 3.3);

the common time controls, the leave-one-out jackknife, and the borderline-classification variants (Section 4.4.2); and the estimator and sensitivity analyses of the subsections that follow.

B.1 Descriptive statistics for the complementary measures

Table 9: Top inheritance rate across the panel, by era

Era	N	Countries	Mean	SD	Min	Max
Pre-war (1929–1938)	184	19	18.45	15.81	0.00	76.00
War (1939–1945)	133	19	28.59	20.81	0.00	77.00
Early postwar (1946–1970)	475	19	37.77	24.15	0.00	90.00
Late twentieth century (1971–2000)	570	19	32.77	22.68	0.00	80.00
Full panel (1929–2000)	1,362	19	32.17	23.06	0.00	90.00

Notes: Top statutory marginal rate on inheritances and estates, from [Scheve and Stasavage \(2012\)](#), for the 19 sample countries with usable estate-tax records. N is the number of country-year observations with a non-missing rate; the series ends in 2000, so no observations fall in 2001–2010. Mean, SD, Min, and Max are in percentage points.

Table 10: Pre-tax and post-tax inequality measures across the panel

Measure	N	Countries	Coverage	Mean	SD	Min	Max
Top-10% share, pre-tax	1,987	34	1929–2010	38.8	10.6	19.0	67.5
Top-10% share, post-tax	1,127	34	~1980–2010	35.6	13.6	17.5	66.0
Top-1% share, pre-tax	1,987	34	1929–2010	13.2	6.0	2.6	35.8
Top-1% share, post-tax	1,127	34	~1980–2010	11.4	6.4	2.3	32.8
Gini, pre-tax	1,987	34	1929–2010	49.7	10.7	28.6	76.5
Gini, post-tax	1,105	34	~1980–2010	43.7	15.8	17.3	74.4

Notes: World Inequality Database income shares and Gini coefficients, pre-tax (`pt inc`) and post-tax (disposable income), for the 34 sample countries with WID series merged into the panel; shares and Gini are on a 0–100 scale. N is the number of non-missing country-year observations; post-tax series are available for most countries only from around 1980. The redistribution wedge analyzed in Section 4 is the pre-tax value minus the post-tax value.

B.2 Additional robustness tables

Table 11 re-estimates equation (1) with one common function of calendar time added and everything else unchanged, so every row reports the same cumulative effect. A post-1980 indicator leaves the

thirty-year estimate at +33.4 percentage points. A linear trend raises it to +39.6 and a quadratic trend lowers it to +25.7: conditional on the lagged rate the fitted common drift is slightly negative, so a straight line tilts the counterfactual downward, while the quadratic bends toward the mid-century rise and fall. Year fixed effects give the lowest value of the family at +11.8, and the five-year effects range from +21.8 to +34.0 and exclude zero in every specification.

Table 11: Cumulative effect under common time controls

Time control	Δ_0	Δ_5	Δ_{30}	Δ_{40}
None (main specification)	+9.1 [+5.2, +13.6]	+32.2 [+16.1, +47.2]	+32.7 [+10.3, +49.9]	+24.0 [−0.7, +43.0]
Post-1980 indicator	+9.2 [+5.3, +13.6]	+32.6 [+16.5, +48.0]	+33.4 [+10.7, +50.4]	+25.2 [+0.6, +43.9]
Linear trend	+9.4 [+5.6, +13.8]	+34.0 [+18.0, +48.8]	+39.6 [+18.8, +55.6]	+32.8 [+9.8, +50.5]
Quadratic trend	+8.7 [+4.7, +13.1]	+29.8 [+13.8, +45.0]	+25.7 [−1.2, +47.4]	+18.1 [−10.7, +42.5]
Year fixed effects	+6.6 [+2.3, +11.4]	+21.8 [+1.5, +41.2]	+11.8 [−20.7, +37.8]	+0.9 [−34.8, +30.0]

Notes: Each row adds the stated common function of calendar time to equation (1), leaving everything else unchanged; $N = 2,437$ in every row. Percentage points, with 95% wild-cluster-bootstrap intervals on country in brackets. All five specifications pass the joint pre-trend test: the centered wild-cluster-bootstrap p -value for the hypothesis that the pre-event coefficients ($\tau = -5$ to -2) are zero lies between 0.76 and 0.93.

Table 12: Leave-one-country-out jackknife: full table

Dropped	Treated?	$\rho+\delta$	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{30} shift
ITA	T	0.897	27.45	28.18	25.70	25.52	25.26	-7.41
AUS	T	0.893	27.71	31.97	27.06	27.38	28.81	-3.86
JPN	T	0.891	32.03	35.37	29.97	29.64	30.71	-1.96
NZL	T	0.891	30.88	33.42	29.71	29.80	31.25	-1.43
CAN	T	0.889	31.01	34.34	29.95	30.06	31.40	-1.27
ZAF	T	0.888	31.32	32.28	30.00	30.25	31.63	-1.05
PRT	C	0.887	31.77	34.16	30.20	30.37	31.78	-0.90
GTM	C	0.887	31.81	34.22	30.27	30.45	31.86	-0.82
VEN	C	0.887	31.88	34.32	30.38	30.57	31.99	-0.69
NOR	C	0.887	32.04	34.55	30.65	30.86	32.30	-0.37
CRI	C	0.887	32.11	34.65	30.77	30.99	32.44	-0.24
BRA	C	0.887	32.16	34.72	30.85	31.08	32.53	-0.15
CHL	C	0.887	32.18	34.76	30.89	31.12	32.58	-0.10
PHL	C	0.887	32.19	34.76	30.90	31.12	32.58	-0.09
PER	C	0.887	32.20	34.79	30.93	31.16	32.62	-0.06
URY	C	0.887	32.23	34.83	30.98	31.21	32.68	0.00
ARG	C	0.887	32.24	34.84	30.99	31.22	32.69	+0.01
MEX	C	0.887	32.24	34.85	30.99	31.23	32.70	+0.02
DNK	C	0.887	32.25	34.86	31.01	31.25	32.72	+0.04
CHE	C	0.887	32.27	34.88	31.04	31.28	32.74	+0.07
BEL	C	0.887	32.30	34.92	31.08	31.32	32.80	+0.12
SWE	T	0.887	32.08	34.76	30.98	31.22	32.81	+0.14
ESP	C	0.887	32.31	34.94	31.10	31.34	32.82	+0.14
AUT	C	0.887	32.34	34.99	31.16	31.41	32.89	+0.21
COL	C	0.887	32.36	35.01	31.18	31.43	32.91	+0.23
IND	C	0.887	32.45	35.14	31.33	31.60	33.09	+0.41
FIN	T	0.883	34.39	34.70	32.03	32.30	33.24	+0.56
NLD	C	0.887	32.58	35.33	31.55	31.83	33.34	+0.66
ISL	C	0.887	32.74	35.56	31.83	32.13	33.66	+0.98
FRA	T	0.883	32.56	36.30	32.07	32.27	33.74	+1.06
KOR	C	0.887	32.87	35.75	32.05	32.36	33.90	+1.23
IRL	T	0.889	34.26	37.37	32.88	33.46	34.98	+2.31
USA	T	0.883	34.04	36.18	31.78	31.92	35.24	+2.56
DEU	T	0.879	35.09	37.49	33.33	33.48	35.51	+2.83
GBR	T	0.882	35.65	39.43	35.89	36.90	38.47	+5.79

Notes: Re-estimate the main specification dropping each of the 35 countries in turn. Sorted by Δ_{30} shift relative to the full-sample main estimate (+32.68). LOO median +32.70; LOO range [+25.26, +38.47].

Table 13: Borderline classification sensitivity: Δ_τ by variant

Variant (n_T)	Δ_0	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{40}	Δ_{50}	Δ_{70}
Main specification (13)	+9.08	+32.23	+34.83	+30.98	+31.21	+32.68	+24.03	+8.90	+2.01
(a) Strict: drop SWE, IRL (11)	+8.85	+35.03	+38.59	+34.30	+35.01	+36.78	+29.09	+13.05	+7.83
(b) Inclusive: add AUT (14)	+8.55	+30.61	+33.41	+30.93	+30.40	+32.47	+25.28	+10.42	+4.04
(c) Combined: drop SWE/IRL, add AUT (12)	+8.37	+33.50	+37.50	+35.04	+34.92	+37.49	+31.47	+15.85	+11.10

Notes: Cumulative Δ_τ in percentage points; the sample is fixed at 35 countries in every variant, and only the war-active/comparison split changes. n_T is the number of war-active countries; Austria enters the inclusive and combined variants with a 1939 entry year. Confidence intervals and pre-trend tests for each variant are reported in Appendix Table 14.

Table 14: Borderline classification: full Δ_τ table with 95% CIs

	Δ_0	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{40}	Δ_{50}	Δ_{70}
(a) Strict (drop SWE, IRL); $n_T = 11$; $\rho + \delta = 0.888$; pre- F $p = 0.97$									
point	+8.85	+35.03	+38.59	+34.30	+35.01	+36.78	+29.09	+13.05	+7.83
CI lo	+4.75	+17.72	+18.68	+14.90	+13.44	+12.36	+3.45	-14.16	-18.39
CI hi	+13.24	+51.28	+54.88	+48.24	+49.83	+53.64	+47.77	+33.21	+26.06
(b) Inclusive (add AUT); $n_T = 14$; $\rho + \delta = 0.898$; pre- F $p = 0.96$									
point	+8.55	+30.61	+33.41	+30.93	+30.40	+32.47	+25.28	+10.42	+4.04
CI lo	+4.74	+14.48	+14.58	+12.61	+10.18	+8.91	-0.09	-14.78	-21.39
CI hi	+12.89	+45.43	+49.49	+44.47	+45.76	+49.62	+44.91	+30.41	+23.90
(c) Combined (drop SWE/IRL, add AUT); $n_T = 12$; $\rho + \delta = 0.898$; pre- F $p = 0.93$									
point	+8.37	+33.50	+37.50	+35.04	+34.92	+37.49	+31.47	+15.85	+11.10
CI lo	+4.46	+16.49	+17.74	+16.23	+13.90	+12.54	+5.88	-12.84	-15.50
CI hi	+12.56	+49.06	+53.43	+48.73	+49.58	+54.98	+51.44	+36.81	+30.99

Notes: Three classification variants of the main 35-country specification (sample fixed at 35 countries; only the treated/control split varies). Wild cluster bootstrap 95% confidence intervals (999 replications, Rademacher weights).

B.3 Heterogeneity-robust estimator for staggered timing

When entry is staggered across cohorts, pooled event-study coefficients can be contaminated by comparisons that use already-treated units as controls. The interaction-weighted estimator of [Sun and Abraham \(2021\)](#) avoids this by forming the effect for each entry cohort separately and then averaging, so no already-treated cohort serves as a control for a later one. The resulting per-period path is almost identical to the pooled one: across the 72 post-entry horizons the two differ by less than one percentage point at 69, with a maximum gap of 1.7 percentage points (Table [15](#)).

Table 15: Sun–Abraham heterogeneity-robust per-period coefficients vs. pooled event study

τ	Pooled (TWFE)	Sun–Abraham	Absolute gap
–5	–1.03	–1.68	0.66
–4	+2.07	+0.16	1.91
–3	–0.28	–0.75	0.48
–2	–1.28	–1.05	0.24
0	+5.70	+5.40	0.30
5	+16.81	+16.18	0.64
10	+10.16	+10.44	0.28
15	+2.32	+1.62	0.70
20	+0.97	+0.31	0.65
30	–2.13	–2.00	0.13
40	–17.90	–18.51	0.62
50	–15.90	–16.47	0.57

Notes: Per-period β_τ from a two-way fixed-effects event study on country and calendar year, without the lagged dependent variable, and from the heterogeneity-robust interaction-weighted estimator of [Sun and Abraham \(2021\)](#) applied to the same specification.

B.4 Sensitivity to parallel-trends violations

A pre-trend test examines only the pre-period; the question it leaves open is how large a violation of parallel trends after entry would have to be before the conclusions changed. The HonestDiD analysis of [Rambachan and Roth \(2023\)](#) answers it with a benchmark: measure candidate violations in units of the largest deviation observed in the pre-period, and ask, for each per-period coefficient, at what multiple of that worst pre-period deviation the coefficient stops being distinct from zero. That multiple is the breakdown value $\bar{M}_{RM}$, reported per horizon in Table 16; $\bar{M}_{RM} = 1$ means the coefficient survives a post-entry violation as large as the worst one seen before the war, and larger is more robust. The entry-year effect is the most robust, $\bar{M}_{RM} = 1.1$: overturning it would take a violation ten percent larger than anything observed in the pre-period. The later per-period coefficients are individually fragile, with breakdown values of 0.2 at three decades and zero at the intermediate horizons. The cumulative trajectory inherits its robustness from the entry-year impulse, which persistence carries forward, rather than from any single later coefficient.

Table 16: HonestDiD per-period breakdown values

τ	$\hat{\beta}_\tau$ (pp)	CR1 95% CI	Breakdown $\bar{M}_{RM}$
0	+9.08	[+4.66, +13.51]	1.1
5	+5.83	[-2.70, +14.35]	0.0
10	+1.73	[-3.64, +7.10]	0.0
15	+2.73	[-0.06, +5.52]	0.0
20	+3.56	[+0.03, +7.08]	0.0
30	+3.73	[+0.36, +7.10]	0.2

Notes: Per-period $\hat{\beta}_\tau$ from the main specification with its CR1 95% confidence interval, and the relative-magnitudes breakdown value $\bar{M}_{RM}$ of [Rambachan and Roth \(2023\)](#): the smallest pre-period-relative violation of parallel trends at which the robust 95% confidence set for β_τ no longer excludes zero.

B.5 Effect by wartime-exposure category

The two-way war-active/non-war-active split used in the main analysis groups together countries with very different wartime experiences. To see whether the rise in top rates was confined to the countries that mobilized most heavily, we divide wartime exposure into three categories: war-active countries (the 13 that sustained at least two percent mobilization: the active combatants together with the high-mobilization neutrals Sweden and Ireland), invaded non-combatants (countries occupied without mobilizing their own populations, [Weinberg, 1994](#)), and non-active participants (countries that contributed below the mobilization threshold or stayed neutral, including neutral countries such as Switzerland that mobilized without engaging in combat). We then compare the average top rate before the war (1929–1938) with the early postwar average (1945–1955) in each (Table 17). All three groups raised top rates, but by markedly different amounts: the war-active countries by about 32 percentage points, invaded non-combatants by about 27, and non-active participants by about 19. The ordering, largest for the most heavily mobilized, is what the mechanism predicts. These are raw pre-versus-post differences rather than the identified event-study effect. Estimating the categories separately bears this out: comparing war-active countries against non-active participants alone reproduces the main result ($\Delta_{30} = +36.7$ against the main $+32.7$, with pre-trends passing), while treating the eight invaded non-combatants as a treated group, with entry at occupation, yields a positive but imprecise path ($\Delta_{30} \approx +18$, nowhere distinct from zero) that fails the joint pre-trend test, so it reads as descriptive rather than causal.

The size of each group’s rise tracks the depth of its exposure to the war. The war-active

countries, which bore the full shock of fighting, financing, and the compression of wealth at once, raised rates the most. The invaded non-combatants, whose wealth the occupation compressed through expropriation, extraction, and collapsing asset values (Scheidel, 2017), but who fielded no mass armies of their own, rose by less. The non-active participants, exposed only to the common forces of the era, rose least, and it is against their path that the main estimate is measured. The gradient is what the paper’s account implies: the war moved top rates wherever it reached, and moved them most where its shock was deepest.

Table 17: Average top marginal rate before and after the war, by wartime-exposure category

Exposure category	Countries	Pre-war mean	Postwar mean	Change
War-active	13	40.2	72.1	+32.0
Invaded non-combatant	8	31.9	58.5	+26.6
Non-active participant	14	11.6	30.2	+18.6

Notes: Unweighted cross-country mean of the top marginal income tax rate (percentage points) in the pre-war period (1929–1938) and the early postwar period (1945–1955), and the change between them, for the three wartime-exposure categories. These are raw group means, not regression-adjusted estimates. The invaded non-combatants are Austria, Belgium, Denmark, Iceland, the Netherlands, Norway, the Philippines and South Korea; the non-active participants are Argentina, Brazil, Chile, Colombia, Costa Rica, Guatemala, India, Mexico, Peru, Portugal, Spain, Switzerland, Uruguay and Venezuela.

C The model: derivations and parametrization

This appendix section derives the equations of Section 5 and reports the calibration and verified steady states.

Bequest rate (the derivation of equation 6)

The bequest rate $\rho = \pi(1 - \theta)/(1 - \pi\theta)$ in the accumulation equation (6), $k_{t+1}^i = (k_t^i)^\theta (\rho \hat{y}_t^i)^{1-\theta}$, is the optimal savings rate of an adult with log utility (3). Let s denote the savings rate, so the financial bequest is $e = s \hat{y}$ and consumption is $c = (1 - s) \hat{y}$. Substituting into (3) and using (6),

$$U_t^i = (1 - \pi) \left[\ln((1 - s)\hat{y}_t^i) - \frac{(\ell_t^i)^\eta}{\eta} \right] + \pi \left[\theta \ln k_t^i + (1 - \theta) \ln(s\hat{y}_t^i) \right].$$

The first-order condition with respect to s is

$$-\frac{1-\pi}{1-s} + \frac{\pi(1-\theta)}{s} = 0 \quad \Longleftrightarrow \quad s = \frac{\pi(1-\theta)}{1-\pi\theta} \equiv \rho.$$

Labor supply (the derivation of $\ell = \chi(1-\tau)^{1/\eta}$)

The labor-supply rule is

$$\ell_t^i = \chi(1-\tau_t)^{1/\eta}, \quad \chi = \left[\frac{\delta(1-\pi\theta)}{1-\pi} \right]^{1/\eta}.$$

It follows from the labor first-order condition with the production function (4) and the tax scheme (5), given $s = \rho$; labor falls in the tax rate with Frisch elasticity $1/(\eta-1)$.

Variance law of motion (the derivation of equation 8)

The law of motion (8) is $\sigma_{t+1}^2 = P(\tau_t)^2 \sigma_t^2 + (1-\theta)^2(1-\tau_t)^2 v^2$. To derive it, take the variance of $\ln k_{t+1}^i$ from (6). Using $\ln \hat{y}_t^i = (1-\tau_t) \ln y_t^i + \tau_t \ln \tilde{y}_t$, the fact that $\ln \tilde{y}_t$ is common across i , and $\ln y_t^i = \ln z_t^i + \alpha \ln k_t^i + \delta \ln \ell_t^i$ with $\ln \ell_t^i$ common across i ,

$$\text{Var}(\ln y_t^i) = \alpha^2 \sigma_t^2 + v^2, \quad \text{Cov}(\ln k_t^i, \ln \hat{y}_t^i) = \alpha(1-\tau_t) \sigma_t^2.$$

Substituting,

$$\sigma_{t+1}^2 = \theta^2 \sigma_t^2 + 2\theta(1-\theta)\alpha(1-\tau_t) \sigma_t^2 + (1-\theta)^2(1-\tau_t)^2(\alpha^2 \sigma_t^2 + v^2),$$

which factors to (8) on collecting σ_t^2 terms.

Political first-order condition (the derivation of equation 11)

The pivotal voter's ex-ante expected utility, taking expectations over z_t^i before voting, is

$$V^{\text{piv}}(\tau; \sigma^2) = \text{const} + (1-\pi\theta) E_z[\ln \hat{y}^{\text{piv}}] - (1-\pi) \frac{\chi^\eta(1-\tau)}{\eta},$$

where the coefficient $(1-\pi\theta)$ on $E_z[\ln \hat{y}^{\text{piv}}]$ collects the bequest weight from (6) and the consumption weight from (3) under the optimal ρ . Substituting the after-tax income (5), the labor first-order condition, and $\ln k^{\text{piv}} = m + \lambda\sigma$, and differentiating in τ , gives the political first-order condition

(11). The deadweight term $(\delta/\eta)\tau/(1-\tau)$ in (11) is assembled from two channels (after dividing out the common factor $1-\pi\theta$): the labor term $(\delta/\eta)\ln(1-\tau)$ carried inside $E_z[\ln \hat{y}^{\text{piv}}]$, with derivative $-(\delta/\eta)/(1-\tau)$, and the disutility term, with derivative $+(\delta/\eta)$; their sum is $-(\delta/\eta)[1/(1-\tau)-1] = -(\delta/\eta)\tau/(1-\tau)$.

Calibration and verified steady states

The figures and the steady states reported in Section 5 use the illustrative calibration

$$\theta = 0.7, \quad \alpha = 0.95, \quad v^2 = 1.0, \quad \lambda = 2.0, \quad \delta/\eta = 0.01.$$

The calibration is not estimated; it is chosen to place the model in the multiple-equilibria regime where the war-shock argument is operative. Solving (11) for τ at each σ^2 and intersecting with the inequality locus (9) returns three steady states (Table 18); the algebra of equations (7)–(11) reproduces these intersections to four decimal places.

Table 18: Verified steady states of the calibrated model			
	Inequality σ^2	Tax rate τ	Stability
Pre-war steady state	1.821	0.030	stable
Unstable threshold	0.068	0.524	unstable
Post-war steady state	0.014	0.746	stable

Notes: Computed from the verified model at the illustrative calibration. The two stable steady states are local attractors under the joint dynamics of (8) and (11); the unstable threshold separates their basins of attraction.